

Asymptotic decomposition in the problem of joined elastic beams

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The analysis of problem of joined elastic beams is presented in comparison with the engineering and asymptotic approaches. Our analysis is based on three-dimensional elasticity theory model and recently developed method of local perturbation (Gaudiello and Kolpakov, 2011), which seems to be an effective tool for analysis of fields in the vicinity of joint. We demonstrate that the method of local perturbation developed in (Gaudiello and Kolpakov, 2011) for scalar Laplace equation can be modified for vectorial elasticity theory problem. We demonstrate that the elasticity theory problem in joined domains of small diameter can be decomposed into one-dimensional problem describing global deformation of a system of joined beams and three-dimensional problems describing local deformation of singular joints in uniform fields. The first problem is the classical one, which ignores individual properties of joint absolutely. The second problem initiates reminiscence about the cellular problem of the homogenization theory for periodic structure. In spite of some similarities, the mentioned problems differ significantly. In particular, the joint of normal type (the joint similar in dimensions and material characteristics to the joined beams) does not manifest itself on global level. Due to the strong localization of perturbation of solution, computation of local strains and stresses in the vicinity of joint can be realized with standard FEM software.

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Introduction

The problem of joint of structural elements traditionally attracted attention of engineers and researchers in structural mechanics, see, e.g., [19, 27] and references in these books. In fact, joints often determine the strength of structures and devices.

A frame – a system of joined beams, is the unity of beams and joints. The problem of deformation of frame allows decomposition: the problem of deformation of beams may be solved independently on the problem of deformation of joints. Conditions in joints have very simple form: conditions of absence of jump for displacements and equilibrium conditions for forces and moments. This model is known as structural mechanics model (the history of this model, related to the names of outstanding scientists and engineers, can be found in [35]). This decomposition has, as usually, two sides. On one hand, with it became possible to analyze frames by using relatively simple mathematical methods (methods of linear algebra and ordinary differential equations in the general case, for specific problems there were developed tricks which allow solve the problems using minimal mathematical knowledge). This (relatively) simplicity in solution of problem of structural mechanics was important element of progress in practical engineering. On the other hand the structural mechanics model absolutely ignores the local structure (both geometrical and material characteristics) of joints and, as a result, provides us with no information about stress-strain state in the vicinity of joints (“vicinity of joint” means the joint itself and the region adjacent to the joint). Stress-state state in the vicinity of joints of general form is very complicated, thus, analysis of joints assumes application of elasticity theory and solution of partial differential equations. Effective analysis of joints became possible after development of numerical analysis and computers programs in the second part of XXth century.

Structural analysis and analysis of joints were “separated” for a time in spite of their natural unity. We do not discuss the engineer methods of analysis of joints as specific object (such kind methods are presented in numerous engineer textbooks, see, e.g. [11, 14, 31]), only note that the discipline treated joints had the name “machine elements” for a long time (in the present names like “design of joints” are used widely, see Blake [7, 8], which is different from the discipline “structural mechanics”).

At the end of XXth century, the problem of joints was attacked with mathematical methods. The most widely used was the asymptotic method, see, e.g., [2, 10, 17, 21, 31, 36] in combination with the methods of functional analysis, see,

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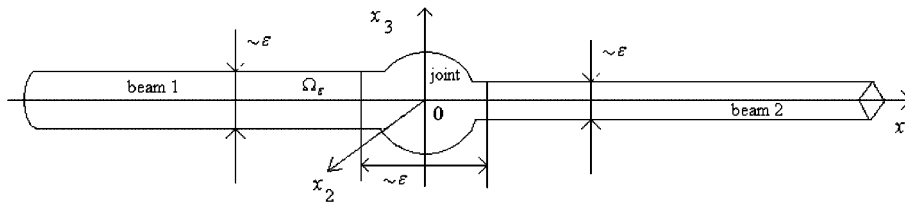


Fig. 1 Two joined beams.

e.g., [9, 12, 16] and the method of boundary layers, see, e.g., [18, 29] (the list of references is not completed). These methods have great theoretical value, but limitation in engineering applications. A serious limitation is assumption that the joint has a simple structure. Even sufficiently general condition from [10] that domain with the notch can be mapped to a cylindrical domain, is restricted for practice. To understand this, it is sufficient to remind the basic types of joints: bolted, riveted and welded, which have structure more complex rather than notched beam (joints can have holes or inclusions). Now, there exists an interest to joint of molecular structures, e.g., nanotubes, which have discrete structure, see, e.g., [28]. These open problems stimulate interest to new methods for analysis of joined plates.

The authors' point of view is that the separation of analysis of frameworks from the analysis of joints was effective approach at the time. Now, it is time to integrate structural mechanics and elasticity theory and analyze frameworks and joints simultaneously. The integrity may be realized on the basis of combination of asymptotic and numerical methods.

The method used in the paper (called the method of local perturbations, see [15]) belongs to the so-called multi-scale methods. The main idea of this method to analyze solitary inhomogeneity in uniform field correlates with the idea homogenization method [3–6, 13] to analyze periodic system of inhomogeneities in uniform field. The difference between two methods is also seen. Since the analysis of inhomogeneities in uniform fields assumes the relatively small size of inhomogeneities, the problem is asymptotic in nature.

Our paper concerns the joints which have dimensions compared with the characteristic diameter of beams under condition that material characteristics of the joint and the beams have the same order. We refer such joints as “normal type joints” to separate them from joints with deep notches [10] and joints on the basis of soft layers [23] (which we referred as degenerated joints).

On brief, our result can be formulated as follows:

1. Individual properties of joints of normal type do not influence global property of framework.
2. A joint demonstrates its specific on the local level. The computation of local strains/stresses in the vicinity of joint is reduced to solution to a special three-dimensional (two-dimensional) elasticity theory problem for an “infinite” beam subjected to basic types (axial tension, bending, and torsion) of overall deformations. Since in the case under consideration strong localization [23] of perturbation takes place, the “infinity” means, in accordance with our numerical experiments, 3–5 diameters of the beam.
3. The problem allows *complete asymptotic decomposition*: solution to the original problem can be separated into solution of a problem about deformation of a system of one-dimensional beams and solution of the elasticity theory problems mentioned in item 2 for particular joints.

In this paper, we consider a simplest case of two joined beams with one joint, because in the general case (the case of a framework formed of beams) the number of varying functions is large and corresponding formulas are complicated. In the case under consideration, formulas are not so complicated and do not mask the ideas of the method of local perturbations as applied to the problem of joined beams.

1 Joined uniform beams

We deal with the body which consists of two long cylinder-like domains (referred in the sequel “beams”) and a domain of arbitrary geometry between those (referred in the sequel “joint”), see Fig. 1. We interpret Ω_ϵ as two beams joined together.

In our consideration the left beam places between $x_1 = -1$ and $x_1 = 0$, the right beam places between $x_1 = 0$ and $x_1 = 1$ (the longitudinal coordinate referred to the length of the beam). The joint places around the origin of the coordinate system $\mathbf{x} = 0$, see Fig. 1. We cannot indicate where the beams are ended and where the joint is began. We only assume that the domain “joint” has sizes in all directions compared with the beam diameters. This assumption is satisfied for typical engineering constructions and devices. In other words, we assume that the diameters of beams and all sizes of the joint are proportional to a small parameter ϵ .

We are expecting that relatively far from the joint the influence of the joints is negligible due to Saint-Venant's principle and beams behave themselves as uniform ones. For this reason, we introduce the notion of the "main part" of a beam as the part of beam placed relatively far from the joint. From the results of our numerical experiments presented below, it will be seen that "relatively far" means distance about 3–5 characteristic diameters of the beams. In the case under consideration, the main parts of beams are cylinders, see Fig. 1, thus, deformation of the main parts of beams can be described in the frameworks of the asymptotic theory of beams (about the asymptotic theory of beams see, e.g., [29, 34]; this theory is based on the asymptotic analysis of solution of elasticity theory problem in small diameter domain and construction of limit model as the diameter tends to zero).

We assume that the beams are clamped at their left and right ends $x_1 = \pm 1$, and subjected to the mass force $\mathbf{f}^\varepsilon = (\varepsilon f, \varepsilon^2 f_2, \varepsilon^2 f_3)(x_1)$. It is assumed that the normal stress $\sigma_{ij}n_j = 0$ on the lateral surface of the beams and joint ($\mathbf{n}$ means the normal vector).

The elasticity problem under consideration can be written in the form of the minimization problem [33]

$$L(\mathbf{u}) = \int_{\Omega_\varepsilon} E(\mathbf{x}, \mathbf{u}) d\mathbf{x} - \int_{\Omega_\varepsilon} \mathbf{f}^\varepsilon(x_1) \mathbf{u}(\mathbf{x}) d\mathbf{x} \rightarrow \min, \mathbf{u} \in V, \quad (1)$$

where

$$E(\mathbf{x}, \mathbf{u}) = \frac{1}{2} \sum_{k,l=1}^3 \sigma_{kl} u_{k,l}^\varepsilon = \frac{1}{2} \sum_{i,j=1}^3 a_{ijkl} u_{k,l}^\varepsilon u_{i,j}^\varepsilon$$

is the local density of the elastic energy; a_{ijkl} are elastic constants (we assume for sake of simplicity that the beams and the joint are made of the same homogeneous material, thus $a_{ijkl} = \text{const}$ in Ω_ε ; $V = \{\mathbf{u} \in \{H^1(\Omega_\varepsilon)\}^n (n = 2, 3) : \mathbf{u}(\mathbf{x}) = 0 \text{ if } x_1 = \pm 1\}$ is the set of admissible displacements; the subscript $, j$ means $\partial/\partial x_j$).

We propose to search solution $\mathbf{u}^\varepsilon(\mathbf{x})$ to problem (1) in the following form (hereafter prime means the differentiating with respect to the variable x_1)

$$\begin{aligned} \mathbf{u}^\varepsilon(\mathbf{x}) = & \varepsilon w_1(x_1) \mathbf{e}_1 + w_2(x_1) \mathbf{e}_2 - w_2'(x_1) x_2 \mathbf{e}_1 + w_3(x_1) \mathbf{e}_3 - w_3'(x_1) x_3 \mathbf{e}_1 \\ & + (x_2 \mathbf{e}_3 - x_3 \mathbf{e}_2) \Theta(x_1) + \varepsilon \mathbf{v}(\mathbf{x}/\varepsilon). \end{aligned} \quad (2)$$

We give comments on the terms in (2). The leading six terms in the right-hand side of (2) describes the behavior of the "main parts" of beams, which are homogeneous cylindrical beams.

The first one corresponds to axial tension of beam, the next fourth terms correspond to bending (in Ox_1x_2 - and Ox_1x_3 -planes, respectively), and the sixth term corresponds to torsion of beam. So, the leading six terms in the right-hand side of (2) correspond to classical hypothesis of engineering theory of beams [33], as well as asymptotic theory of the asymptotic theory for uniform beams [20, 34]. We accept these terms with no additional discussion.

The novelty is the last term in the right-hand side of (2). It is introduced to take into account the perturbation of solution associated with the joint. We assume that the function $\mathbf{v}(\mathbf{y})$ is perturbed in the vicinity of joint. The exact meaning on the term "perturbation" for the structure under consideration will be specified later.

The derivatives of the displacements (2) are (we denote by $\mathbf{y} = \frac{\mathbf{x}}{\varepsilon}$ the "fast" variables and remind the differentiating rule $\partial/\partial x_i = \frac{1}{\varepsilon} \partial/\partial y_i$ when one differentiates function $\mathbf{v}(\mathbf{x}/\varepsilon)$ of the "fast" variables, [32])

$$\begin{aligned} u_{k,l}^\varepsilon(\mathbf{x}) = & \varepsilon \delta_{k1} w_1'(x_1) \delta_{l1} + \delta_{k2} w_2'(x_1) \delta_{l1} - \delta_{k1} w_2'(x_1) \delta_{l2} - \varepsilon y_2 \delta_{k1} w_2''(x_1) \delta_{l1} \\ & + \delta_{k3} w_3'(x_1) \delta_{l1} - \delta_{k1} w_3'(x_1) \delta_{l3} - \varepsilon y_3 \delta_{k1} w_3''(x_1) \delta_{l1} \\ & + (\delta_{l2} \delta_{k3} - \delta_{l3} \delta_{k2}) \Theta(x_1) + \varepsilon (y_2 \delta_{k3} - y_3 \delta_{k2}) \Theta'(x_1) \delta_{l1} + v_{k,ly}(\mathbf{y}). \end{aligned} \quad (3)$$

Hereafter, the subscript $, jy$ means $\partial/\partial y_j$.

The local stresses $\sigma_{ij} = a_{ijkl} u_{k,l}^\varepsilon(\mathbf{x})$ corresponding to the displacements (2) are

$$\begin{aligned} \sigma_{ij} = & \varepsilon a_{ij11} w_1'(x_1) + a_{ij21} w_2'(x_1) - a_{ij12} w_2'(x_1) - \varepsilon y_2 a_{ij11} w_2''(x_1) \\ & + a_{ij31} w_3'(x_1) - a_{ij13} w_3'(x_1) - \varepsilon y_3 a_{ij11} w_3''(x_1) \\ & + (a_{ij32} - a_{ij23}) \Theta(x_1) + (y_2 a_{ij31} - y_3 a_{ij21}) \Theta'(x_1) + a_{ijkl} v_{k,ly}(\mathbf{y}) \\ = & \varepsilon a_{ij11} w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{ij11} w_\alpha''(x_1) + \varepsilon (y_2 a_{ij31} - y_3 a_{ij21}) \Theta'(x_1) + a_{ijkl} v_{k,ly}(\mathbf{y}). \end{aligned} \quad (4)$$

In Eqs. (4) we use the symmetry of elastic constants: $a_{ijkl} = a_{ijlk}$ [33].

The local energy $E(\mathbf{x}, \mathbf{u}^\varepsilon) = \frac{1}{2} \sigma_{kl} u_{k,l}^\varepsilon(\mathbf{x})$ corresponding to the displacements $\mathbf{u}^\varepsilon(\mathbf{x})$ (2) is

$$\begin{aligned} E(\mathbf{x}, \mathbf{u}^\varepsilon) &= \frac{1}{2} \sigma_{kl} \times [\varepsilon \delta_{k1} w_1'(x_1) \delta_{l1} + \delta_{k2} w_2'(x_1) \delta_{l1} - \delta_{k1} w_2'(x_1) \delta_{l2} - \varepsilon y_2 \delta_{k1} w_2''(x_1) \delta_{l1} \\ &= \delta_{k3} w_3'(x_1) \delta_{l1} - \delta_{k1} w_3'(x_1) \delta_{l3} - \varepsilon y_3 \delta_{k1} w_3''(x_1) \delta_{l1} \\ &\quad + (\delta_{l2} \delta_{k3} - \delta_{l3} \delta_{k2}) \Theta(x_1) + \varepsilon (y_2 \delta_{k3} - y_3 \delta_{k2}) \Theta'(x_1) \delta_{l1} + v_{k,ly}(\mathbf{y})] \\ &= \frac{\varepsilon}{2} \sigma_{11} w_1'(x_1) + \frac{1}{2} [(\sigma_{21} - \sigma_{12}) w_2'(x_1) - \sigma_{11} \varepsilon y_2 w_2''(x_1) \\ &\quad + (\sigma_{31} - \sigma_{13}) w_3'(x_1) - \sigma_{11} \varepsilon y_3 w_3''(x_1) + (\sigma_{32} - \sigma_{23}) \Theta(x_1) \\ &\quad + (\sigma_{31} y_2 - \sigma_{21} y_3) \Theta'(x_1) + \sigma_{kl} v_{k,ly}(\mathbf{y})] \\ &= \frac{\varepsilon}{2} \sigma_{11} w_1'(x_1) + \frac{1}{2} \left[-\sigma_{11} \varepsilon \sum_{\alpha=2,3} y_\alpha w_\alpha''(x_1) + \varepsilon (\sigma_{31} y_2 - \sigma_{21} y_3) \Theta'(x_1) + \sigma_{kl} v_{k,ly}(\mathbf{y}) \right]. \end{aligned} \quad (5)$$

In Eqs. (5), we use formula (3) and the symmetry of the stress tensor $\sigma_{ij} = \sigma_{ji}$. We see that all the terms corresponding to the engineering theory in the right-hand side of (5) have the same order. It is a result of the scaling accepted in (2).

For sake of simplicity, we consider the beams and the joint made of isotropic material for which $a_{1213} = a_{1312} = a_{1121} = a_{1131} = 0$ [33]. For this case, substituting stresses σ_{ij} (4) into Eqs. (5), we obtain

$$\begin{aligned} E(\mathbf{x}, \mathbf{u}^\varepsilon) &= \frac{1}{2} \left[\varepsilon a_{1111} w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{1111} w_\alpha''(x_1) + a_{11kl} v_{k,ly}(\mathbf{y}) \right] \varepsilon w_1'(x_1) \\ &\quad - \frac{1}{2} \sum_{\beta=2,3} \left[\varepsilon a_{1111} w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{1111} w_\alpha''(x_1) + a_{11kl} v_{k,ly}(\mathbf{y}) \right] \varepsilon y_\beta w_\beta''(x_1) \\ &\quad + \frac{1}{2} \varepsilon^2 (y_2^2 a_{3131} + y_3^2 a_{2121}) \Theta'^2(x_1) \\ &\quad + \frac{1}{2} \left[\varepsilon a_{ij11} w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{ij11} w_\alpha''(x_1) + \varepsilon (y_2 a_{ij31} - y_3 a_{ij21}) \Theta'(x_1) \right. \\ &\quad \left. + a_{ijkl} v_{k,ly}(\mathbf{y}) \right] v_{i,jy}(\mathbf{y}). \end{aligned} \quad (6)$$

As a result, we arrive at the following minimization problem:

$$\begin{aligned} L(w_1, w_2, w_3, \Theta, \mathbf{v}) &= \int_{\Omega_\varepsilon} E(\mathbf{x}, \mathbf{u}^\varepsilon) d\mathbf{x} \\ &\quad - \int_{\Omega_\varepsilon} \mathbf{f}^\varepsilon(x_1) [\varepsilon w_1(x_1) \mathbf{e}_1 + w_2(x_1) \mathbf{e}_2 - w_2'(x_1) \varepsilon y_2 \mathbf{e}_1 + w_3(x_1) \mathbf{e}_3 - w_3'(x_1) \varepsilon y_3 \mathbf{e}_1 \\ &\quad + \varepsilon (y_2 \mathbf{e}_3 - y_3 \mathbf{e}_2) \Theta(x_1) + \varepsilon \mathbf{v}(\mathbf{x}/\varepsilon)] d\mathbf{x} \rightarrow \min; \end{aligned} \quad (7)$$

$$w_1, \Theta \in H^1[-1, 1]; \quad w_2, w_3 \in H^2[-1, 1]; \quad \mathbf{v} \in V_{loc};$$

where $E(\mathbf{x}, \mathbf{u}^\varepsilon)$ is given by the expression (6) and $V_{loc} = \{\mathbf{v} \in V : \mathbf{v}(\mathbf{x}) \rightarrow 0 \text{ as } |\mathbf{x}| \rightarrow \infty\}$.

Functional $L(w_1, w_2, w_3, \Theta, \mathbf{v})$ depends on five functional variables: four functions $w_1(x_1)$, $w_2(x_1)$, $w_3(x_1)$, $\Theta(x_1)$ of one “slow” variable x_1 and function $\mathbf{v}(\mathbf{y})$ of “fast” variables $\mathbf{y}$. Euler-Lagrange equations for the functional $L(w_1, w_2, w_3, \Theta, \mathbf{v})$ are the following (δL means variation of functional with respect to the corresponding function)

$$\begin{aligned} \delta L_{\mathbf{v}} = & \int_{\Omega_\varepsilon} [\varepsilon a_{11kl} w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl} w''_\alpha(x_1) + \frac{1}{2} \varepsilon (y_2 a_{kl31} - y_3 a_{kl21}) \Theta'(x_1) \\ & + a_{ijkl} v_{i,jy}(\mathbf{y})] \delta v_{k,ly}(\mathbf{y}) d\mathbf{x} - \int_{\Omega_\varepsilon} (\varepsilon^2 f_1(x_1) \delta v_1^\varepsilon + \varepsilon^3 \sum_{\alpha=2,3} f_\alpha(x_1) \delta v_\alpha^\varepsilon) d\mathbf{x} = 0, \end{aligned} \quad (8)$$

and

$$\begin{aligned} \delta L_{w_1} = & \int_{\Omega_\varepsilon} [\varepsilon a_{1111} w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{1111} w''_\alpha(x_1) + a_{11kl} v_{k,ly}(\mathbf{y})] \varepsilon \delta w'_1(x_1) d\mathbf{x} \\ & - \int_{\Omega_\varepsilon} \varepsilon^2 f_1(x_1) \delta w_1(x_1) d\mathbf{x} = 0, \end{aligned} \quad (9)$$

$$\begin{aligned} \delta L_{w_\alpha} = & \int_{\Omega_\varepsilon} [\varepsilon a_{1111} w'_1(x_1) - \varepsilon \sum_{\beta=2,3} y_\beta a_{1111} w''_\beta(x_1) + a_{11kl} v_{k,ly}(\mathbf{y})] \varepsilon y_\alpha \delta w''_\alpha(x_1) d\mathbf{x} \\ & - \int_{\Omega_\varepsilon} [\varepsilon^2 f_\alpha(x_1) \delta w_2(x_1) - \varepsilon^2 f_1(x_1) \delta w'_\alpha(x_1) y_2] d\mathbf{x} = 0 \quad (\alpha = 2, 3), \end{aligned} \quad (10)$$

$$\begin{aligned} \delta L_\Theta = & \int_{\Omega_\varepsilon} [\varepsilon^2 (y_2^2 a_{3131} + y_3^2 a_{2121}) \Theta'(x_1) + \frac{1}{2} \varepsilon (y_2 a_{ij31} - y_3 a_{ij21}) v_{i,jy}(\mathbf{y})] \delta \Theta'(x_1) d\mathbf{x} \\ & - \int_{\Omega_\varepsilon} \varepsilon^3 [f_3(x_1) y_2 - f_2(x_1) y_3] \Theta(x_1) d\mathbf{x}. \end{aligned} \quad (11)$$

In Eq. (8), the leading terms have orders of ε . We omit the terms of order higher than ε in Eq. (8). Integrating the remaining terms in Eq. (8) by parts, we arrive at the following boundary-value problem:

$$\begin{aligned} [a_{ijkl} v_{i,jy} + \varepsilon a_{11kl} w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl} w''_\alpha(x_1) + \frac{1}{2} \varepsilon (y_2 a_{kl31} - y_3 a_{kl21}) \Theta'(x_1)]_{,ly} &= 0 \quad \text{in } \Omega_\varepsilon, \\ [a_{ijkl} v_{i,jy} + \varepsilon a_{11kl} w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl} w''_\alpha(x_1) + \frac{1}{2} \varepsilon (y_2 a_{kl31} - y_3 a_{kl21}) \Theta'(x_1)] n_l &= 0 \\ \text{on the lateral side of } \Omega_\varepsilon, & \\ \mathbf{v}(\mathbf{x}/\varepsilon) \text{ is perturbed near the point } \mathbf{y} = 0. & \end{aligned} \quad (12)$$

Separating in (12) “slow” and “fast” variables (the separation procedure assumes that problem (12) is treated as a problem in the “fast” variables $\mathbf{y}$, in addition functions of the “slow” variable x_1 in (12) are considered as parameters [32], we can write solution to Eq. (12) in the form

$$\mathbf{v}(\mathbf{x}/\varepsilon) = \varepsilon \mathbf{N}^{(1)}(\mathbf{x}/\varepsilon) w'_1(x_1) + \varepsilon \sum_{\alpha=2,3} \mathbf{N}^{(\alpha)}(\mathbf{x}/\varepsilon) w''_\alpha(x_1) + \varepsilon \mathbf{M}(\mathbf{x}/\varepsilon) \Theta'(x_1), \quad (13)$$

where function $\mathbf{N}^{(1)}(\mathbf{y})$ is solution to the following boundary value problem in the domain $\Omega_\infty(\varepsilon) = \frac{1}{\varepsilon} \Omega_\varepsilon$:

$$\begin{aligned} [a_{ijkl} N_{i,jy}^{(1)}(\mathbf{y}) + a_{11kl}]_{,ly} &= 0 \quad \text{in } \Omega_\infty(\varepsilon), \\ [a_{ijkl} N_{i,jy}^{(1)}(\mathbf{y}) + a_{11kl}] n_l &= 0 \quad \text{on lateral side of } \Omega_\infty(\varepsilon), \\ \mathbf{N}^{(1)}(\mathbf{y}) \text{ is perturbed near the point } \mathbf{y} = 0, & \end{aligned} \quad (14)$$

corresponding to axial tension of beam.

Function $\mathbf{N}^{(\alpha)}(\mathbf{y})$ ($\alpha = 2, 3$ is solution to the problem

$$\begin{aligned} [a_{ijkl}N_{i,jy}^{(\alpha)}(\mathbf{y}) - y_{\alpha}a_{11kl}]_{,ly} &= 0 \quad \text{in } \Omega_{\infty}(\varepsilon), \\ [a_{ijkl}N_{i,jy}^{(\alpha)}(\mathbf{y}) - y_{\alpha}a_{11kl}]n_l &= 0 \quad \text{on lateral side of } \Omega_{\infty}(\varepsilon), \\ \mathbf{N}^{(\alpha)}(\mathbf{y}) &\text{ is perturbed near the point } \mathbf{y} = 0, \end{aligned} \quad (15)$$

corresponding to bending of beam.

Function $\mathbf{M}(\mathbf{y})$ is solution to the boundary value problem

$$\begin{aligned} [a_{ijkl}M_{i,jy}(\mathbf{y}) - \frac{1}{2}(y_2a_{kl31} - y_3a_{kl21})]_{,ly} &= 0 \quad \text{in } \Omega_{\infty}(\varepsilon), \\ [a_{ijkl}M_{i,jy}(\mathbf{y}) - \frac{1}{2}(y_2a_{kl31} - y_3a_{kl21})]n_l &= 0 \quad \text{on lateral side of } \Omega_{\infty}(\varepsilon), \\ \mathbf{M}(\mathbf{y}) &\text{ is perturbed near the point } \mathbf{y} = 0, \end{aligned} \quad (16)$$

corresponding to torsion of beam.

We now discuss the condition “is perturbed near the point $\mathbf{y} = 0$ ” in the problems above. We note, that solutions to the problems (14)–(16) in cylindrical domains are non zero. It is a specific of the elasticity theory problem (in scalar problems such kind problem does not arise, [15]). We note that the first coordinate of solutions the problems (14)–(16) in cylindrical domains are zero and the second and the third coordinates depend on the variables x_2 and x_3 , only. In other words, solutions of problems (14)–(16) in cylindrical domain have the form $\mathbf{N}_0^{(i)}(y_2, y_3) = (0, N_{02}^{(i)}, N_{03}^{(i)})(y_2, y_3)$ ($i = 1, 2, 3$) and $\mathbf{M}_0(y_2, y_3) = (0, M_{02}, M_{03})(y_2, y_3)$ (the explicit formulas for $\mathbf{N}_0^{(i)}(y_2, y_3)$ and $\mathbf{M}_0(y_2, y_3)$ were obtained in [24], see Appendix below). These functions describe local transversal deformations of the beam. Such kind deformations absent in the Euler-Bernoulli theory of tension and bending of beam (and leads to contradictions in the engineering theories of beams, namely, nonzero normal stress on the free boundary of beam) and present in the Saint-Venant theory of torsion of beams. The asymptotic theories of uniform beams [20, 34] predict the existence of transversal deformations for all types of deformation of beams. The condition that solution “is perturbed near the point $\mathbf{y} = 0$ ” in the problems (14)–(16) means that $\mathbf{N}^{(i)}(\mathbf{y}) \rightarrow \mathbf{N}_0^{(i)}(y_2, y_3)$ ($i = 1, 2, 3$) and $\mathbf{M}(\mathbf{y}) \rightarrow \mathbf{M}_0(y_2, y_3)$, as $|y_1| \rightarrow \infty$. Correspondingly, the functions $\mathbf{n}^{(i)}(\mathbf{y}) = \mathbf{N}^{(i)}(\mathbf{y}) - \mathbf{N}_0^{(i)}(y_2, y_3)$ ($i = 1, 2, 3$) and $\mathbf{m}(\mathbf{y}) = \mathbf{M}(\mathbf{y}) - \mathbf{M}_0(y_2, y_3)$ are localized near the point $\mathbf{y} = 0$ in the terms of the paper [15].

Substituting (13) into Eq. (9), we have after the grouping collecting of terms

$$\begin{aligned} \int_{\Omega_{\varepsilon}} \left\{ \varepsilon^2 [a_{1111} + a_{11kl}N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)]w_1'(x_1) + \varepsilon^2 \sum_{\alpha=2,3} [-y_{\alpha}a_{1111} + a_{11kl}N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)]w_{\alpha}''(x_1) \right. \\ \left. + \varepsilon^2 a_{11kl}M_{k,ly}(\mathbf{x}/\varepsilon)\Theta'(x_1) \right\} \delta w_1'(x_1) d\mathbf{x} - \varepsilon^2 \int_{\Omega_{\varepsilon}} f_1(x_1) \delta w_1(x_1) d\mathbf{x} = 0. \end{aligned} \quad (17)$$

As was noted above, we normalized the mass forces in such a way, that the tensional deformation has the order of bending and torsional deformations. Thus, the total energy of the system of joined beams (in particular, the left-hand part of Eq. (17)) has the order of ε^{-4} . To obtain from Eq. (17) a nontrivial equation, as $\varepsilon \rightarrow 0$, we have to divide Eq. (17) by ε^{-4} . As a result, we have

$$\begin{aligned} \frac{1}{\varepsilon^2} \int_{\Omega_{\varepsilon}} \left\{ [a_{1111} + N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)]w_1'(x_1) + \sum_{\alpha=2,3} [-y_{\alpha}a_{1111} + a_{11kl}N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)]w_{\alpha}''(x_1) \right. \\ \left. + a_{11kl}M_{k,ly}(\mathbf{x}/\varepsilon)\Theta'(x_1) \right\} \delta w_1'(x_1) d\mathbf{x} + \frac{1}{\varepsilon^2} \int_{\Omega_{\varepsilon}} f_1(x_1) \delta w_1(x_1) d\mathbf{x} = 0. \end{aligned} \quad (18)$$

Substituting Ansatz (13) into (10, $\alpha = 2$), we obtain after the collecting the terms

$$\begin{aligned} & \int_{\Omega_\varepsilon} \left\{ \varepsilon^2 y_2 [a_{1111} + y_2 a_{11kl} N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) \right. \\ & + \varepsilon^2 [-y_2^2 a_{1111} + y_2 a_{11kl} N_{k,ly}^{(2)}(\mathbf{x}/\varepsilon)] w_2''(x_1) + \varepsilon^2 [-y_2 y_3 a_{1111} + y_2 a_{11kl} N_{k,ly}^{(3)}(\mathbf{x}/\varepsilon)] w_3''(x_1) \\ & \left. + \varepsilon^2 y_2 a_{11kl} M_{k,ly}(\mathbf{x}/\varepsilon) \Theta(x_1) \right\} \delta w_2''(x_1) d\mathbf{x} - \varepsilon^2 \int_{\Omega_\varepsilon} [f_2(x_1) \delta w_2(x_1) - f_1(x_1) y_2 \delta w_2'(x_1)] d\mathbf{x} = 0. \end{aligned} \quad (19)$$

Dividing all terms in Eq. (19) by ε^{-4} (the reason for the dividing (19) by ε^{-4} is as above), we obtain

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ y_2 [a_{1111} + a_{11kl} N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) \right. \\ & + [-y_2^2 a_{1111} + y_2 a_{11kl} N_{k,ly}^{(2)}(\mathbf{x}/\varepsilon)] w_2''(x_1) + [y_2 y_3 a_{1111} + y_2 a_{11kl} N_{k,ly}^{(3)}(\mathbf{x}/\varepsilon)] w_3''(x_1) \\ & \left. + y_2 a_{11kl} M_{k,ly}(\mathbf{x}/\varepsilon) \Theta'(x_1) \right\} \delta w_2''(x_1) d\mathbf{x} - \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} [f_2(x_1) \delta w_2(x_1) - f_1(x_1) y_2 \delta w_2'(x_1)] d\mathbf{x} = 0. \end{aligned} \quad (20)$$

Substituting Ansatz (13) into Eq. (10) for $\alpha = 3$ and dividing by ε^{-4} , we obtain in the similar way

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ y_3 [a_{1111} + a_{11kl} N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) \right. \\ & + [-y_2 y_3 a_{1111} + \frac{1}{2} y_2 a_{11kl} N_{k,ly}^{(2)}(\mathbf{x}/\varepsilon)] w_2''(x_1) + [-y_3^2 a_{1111} + y_3 a_{11kl} N_{k,ly}^{(3)}(\mathbf{x}/\varepsilon)] w_3''(x_1) \\ & \left. + y_3 a_{11kl} M_{k,ly}(\mathbf{x}/\varepsilon) \Theta'(x_1) \right\} \delta w_3''(x_1) d\mathbf{x} - \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} [f_3(x_1) \delta w_3(x_1) - f_1(x_1) y_3 \delta w_3'(x_1)] d\mathbf{x} = 0. \end{aligned} \quad (21)$$

Substituting Ansatz (13) into Eq. (11) and dividing by ε^{-4} , we have

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ [(y_2^2 a_{3131} + y_3^2 a_{2121}) + \frac{1}{2} (y_2 a_{ij31} - y_3 a_{ij21}) M_{i,jy}(\mathbf{x}/\varepsilon)] \Theta'(x_1) \right. \\ & + \frac{1}{2} (y_2 a_{ij31} - y_3 a_{ij21}) N_{i,jy}^{(1)}(\mathbf{x}/\varepsilon) w_1'(x_1) \\ & \left. + \frac{1}{2} \sum_{\alpha=2,3} (y_2 a_{ij31} - y_3 a_{ij21}) N_{i,jy}^{(\alpha)}(\mathbf{x}/\varepsilon) w_\alpha''(x_1) \right\} \delta \Theta'(x_1) d\mathbf{x} = 0. \end{aligned} \quad (22)$$

We assume that the function $\nabla \mathbf{n}^{(i)}(\mathbf{y}) = \nabla \mathbf{N}^{(i)}(\mathbf{y}) - \nabla \mathbf{N}_0^{(i)}(y_2, y_3) \in L_1(\Omega_\infty)$ ($i = 1, 2, 3$), where Ω_∞ means domain $\Omega_\infty(\varepsilon)$ infinitely extended along Ox_1 -axis in both directions. Under this assumption $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x} \rightarrow 0$, as $\varepsilon \rightarrow 0$. Actually,

$$\left| \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x} \right| \leq \frac{1}{\varepsilon^2} \varepsilon^3 \int_{\Omega_\infty(\varepsilon)} |a_{11kl} n_{k,ly}^{(i)}(\mathbf{y})| d\mathbf{y} \leq \varepsilon \int_{\Omega_\infty} |a_{11kl} n_{k,ly}^{(i)}(\mathbf{y})| d\mathbf{y} \rightarrow 0, \quad (23)$$

as $\varepsilon \rightarrow 0$.

The expression $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} y_\beta a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x} \rightarrow 0$, as $\varepsilon \rightarrow 0$ ($\beta = 1, 2; i = 1, 2, 3$). Actually,

$$\begin{aligned} \left| \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} y_\beta a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x} \right| &= \left| \frac{1}{\varepsilon^3} \int_{\Omega_\varepsilon} x_\beta a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x} \right| \\ &= \left| \varepsilon \int_{\Omega_\infty(\varepsilon)} y_\beta a_{11kl} n_{k,ly}^{(i)}(\mathbf{y}/\varepsilon) d\mathbf{y} \right| \leq \varepsilon C \int_{\Omega_\infty} |a_{11kl} n_{k,ly}^{(i)}(\mathbf{y})| d\mathbf{y} \rightarrow 0, \end{aligned} \quad (24)$$

as $\varepsilon \rightarrow 0$, where the constant C (thus, $|y_\beta| \leq C \leq \infty$). For the similar reason, $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} a_{11kl} m_{k,ly}(\mathbf{x}/\varepsilon) d\mathbf{x} \rightarrow 0$ and $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} y_\beta a_{11kl} m_{k,ly}(\mathbf{x}/\varepsilon) d\mathbf{x} \rightarrow 0$, as $\varepsilon \rightarrow 0$ ($\beta = 1, 2$), under the assumption that $\nabla \mathbf{m}(\mathbf{y}) = \nabla \mathbf{M}(\mathbf{y}) - \nabla \mathbf{M}_0(y_2, y_3) \in L_1(Q_\infty)$. Thus, as $\varepsilon \rightarrow 0$, we can change in Eqs. (18), (20), (21), and (22) functions $\mathbf{N}^{(i)}(\mathbf{y})$ and $\mathbf{M}(\mathbf{y})$ for $\mathbf{N}_0^{(i)}(y_2, y_3)$ and $\mathbf{M}_0(y_2, y_3)$, respectively. After this change, we can carry out integration in Eqs. (18), (20), (21), and (22) with respect to the variables y_2 and y_3 in explicit form as it is done in asymptotic theories of beams [20, 34] (we remind that the explicit forms of the functions $\mathbf{N}_0^{(i)}(y_2, y_3)$ and $\mathbf{M}_0(y_2, y_3)$ are known for cylindrical beams, see Appendix). As a result, we obtain the following limit equations:

$$\begin{aligned} \int_{-1}^1 \{D_{11}(x_1)w_1' + \frac{1}{2} \sum_{\alpha=2,3} D_{1\alpha}(x_1)w_\alpha'' + F_1(x_1)\Theta'\} \delta w_1'(x_1) dx - \int_{-1}^1 F_1(x_1) \delta w_1 dx &= 0, \\ \int_{-1}^1 \{D_{\alpha 1}(x_1)w_1' + \sum_{\beta=2,3} D_{\alpha\beta}(x_1)w_\beta'' + F_\alpha(x_1)\Theta'\} \delta w_\alpha'' dx & \\ - \int_{-1}^1 [F_\alpha(x_1) \delta w_3 + m_\alpha(x_1) \delta w_\alpha'] dx &= 0 \quad (\alpha = 2, 3), \\ \int_{-1}^1 \{I(x_1)\Theta' + J_1(x_1)w_1'' - \sum_{\alpha=2,3} J_\alpha(x_1)w_\alpha''\} \delta \Theta' dx &= 0. \end{aligned} \quad (25)$$

Here $F_i(x_1) = \int_S f_i(x_1) dy_2 dy_3$ and $m_\alpha(x_1) = \int_S f_1(x_1) y_\alpha dy_2 dy_3$; S means the cross-section of the beam.

Generally, coefficients in Eqs. (20) are different for $x_1 < 0$ and $x_1 > 0$. Integrating the Eqs. (20) by parts, we arrive at the classical equations of tension-bending-torsion of beam and the classical conjugation conditions

$$\begin{aligned} [D_{11}(x_1)w_1' + \sum_{\alpha=2,3} D_{1\alpha}(x_1)w_\alpha'' + F_1(x_1)\Theta'] &= 0 \quad (i = 1, 2, 3), \\ [(D_{\alpha 1}(x_1)w_1' + \sum_{\beta=2,3} D_{\alpha\beta}(x_1)w_\beta'' + F_\alpha(x_1)\Theta')] &= 0 \quad (\alpha = 2, 3), \\ [I(x_1)\Theta' + J_2(x_1)w_1'' + \sum_{\alpha=2,3} J_\alpha(x_1)w_\alpha''] &= 0. \end{aligned} \quad (26)$$

In Eqs. (26), $[\]$ means the “jump” of the corresponding function at the point $x_1 = 0$. It is the fact of interest that although the integrands (the local stresses) in the engineering (Euler–Bernoulli) theory [33] and the asymptotic theories [20, 34] are different (for example, $D_{11} = \int_S a_{1111} dy_2 dy_3$ in the engineering theory and $D_{11} = \int_S (a_{1111} + a_{11kl} N_{0kl}^{(1)}(y_2, y_3)) dy_2 dy_3$ in our computations as well as in the asymptotic theory of beams), the final expressions for the computation of the stiffnesses are the same. Then coefficients D_{ij}, F_i, J_i, I in Eqs. (25) and (26) are computed in accordance with the classical

formulas: $D_{11} = a_{1111} \int_S dy_2 dy_3$ (axial tension/compression stiffness), $D_{\alpha\beta} = a_{1111} \int_S y_\alpha y_\beta dy_2 dy_3$ ($\alpha, \beta = 2, 3$) (bending stiffnesses), and $I = \int_S (y_2^2 a_{3131} + y_3^2 a_{2121}) dy_2 dy_3$ (torsion stiffness). The coefficients $D_{1\alpha}$, $D_{\alpha 1}$ are coupled (sometimes call asymmetrical) coefficient accounting interaction of various types of deformation modes. Note that all coefficients in Eqs. (25) and (26) do not depend on the geometry of the joint and material the joint is made of. It means that the individual properties of joints of normal type do not manifest themselves on the global level.

In accordance with Eq. (4), the local stresses are

$$\begin{aligned} \sigma_{ij} = & [a_{ij11} + a_{ijkl} N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) + \sum_{\alpha=2,3} [-\varepsilon y_\alpha a_{ij11} + a_{ijkl} N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(x_1) \\ & + [\varepsilon(y_2 a_{ij31} - y_3 a_{ij21}) + a_{ijkl} M_{k,ly}(\mathbf{x}/\varepsilon)] \Theta'(x_1). \end{aligned} \quad (27)$$

Introducing the tensors of local stress concentrations

$$K_{ij}^1(\mathbf{y}) = a_{ij11} + a_{ijkl} N_{k,ly}^{(1)}(\mathbf{y}), \quad K_{ij}^\alpha(\mathbf{y}) = -\varepsilon y_\alpha a_{ij11} + a_{ijkl} N_{k,ly}^{(\alpha)}(\mathbf{y}) \quad (\alpha = 2, 3),$$

$$k_{ij}(\mathbf{y}) = \varepsilon(y_2 a_{ij31} - y_3 a_{ij21}) + a_{ijkl} M_{k,ly}(\mathbf{x}/\varepsilon),$$

we can rewrite Eqs. (27) in the form of linear combination of the tensors K_{ij}^1, K_{ij}^α ($\alpha = 2, 3$), k_{ij} , which depend on the variable $\mathbf{x}/\varepsilon$, and functions $w_1', w_\alpha'', \Theta'$ of the “slow” variable x_1 :

$$\sigma_{ij} = K_{ij}^1(\mathbf{x}/\varepsilon) w_1'(x_1) + \sum_{\alpha=2,3} K_{ij}^\alpha(\mathbf{x}/\varepsilon) w_\alpha''(x_1) + k_{ij}(\mathbf{x}/\varepsilon) \Theta'(x_1). \quad (28)$$

In the vicinity of the joint (i.e. near the point $\mathbf{x} = 0$), we can accept the following approximations: $w_1'(x_1) \approx w_1'(0)$, $w_\alpha''(x_1) \approx w_\alpha''(0)$ ($\alpha = 2, 3$), $\Theta(x_1) \approx \Theta(0)$ for the functions of “slow” variable x_1 and write the corresponding approximation for expressions (28)

$$\begin{aligned} \sigma_{ij}(\mathbf{x}/\varepsilon) \approx & [a_{ij11} + a_{ijkl} N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(0) + \sum_{\alpha=2,3} [-\varepsilon y_\alpha a_{ij11} + a_{ijkl} N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(0) \\ & + [\varepsilon(y_2 a_{ij31} - y_3 a_{ij21}) + a_{ijkl} M_{k,ly}(\mathbf{x}/\varepsilon)] \Theta'(0). \end{aligned} \quad (29)$$

We can rewrite this last formula as

$$\sigma_{ij}(\mathbf{x}/\varepsilon) \approx K_{ij}^1(\mathbf{x}/\varepsilon) w_1'(0) + \sum_{\alpha=2,3} K_{ij}^\alpha(\mathbf{x}/\varepsilon) w_\alpha''(0) + k_{ij}(\mathbf{x}/\varepsilon) \Theta'(0).$$

It is seen from the formula (29) that the local stresses in the vicinity of joint depend on the functions $\mathbf{N}^{(i)}(\mathbf{y})$ ($i = 1, 2, 3$) and $\mathbf{M}(\mathbf{y})$, which are determined by geometry and material properties of specific joint (see problems (13) and (15), which introduce these functions).

In the main part of the beams $\mathbf{N}^{(i)}(\mathbf{x}/\varepsilon) \approx \mathbf{N}_0^{(i)}(y_2/\varepsilon, y_3/\varepsilon)$ and $\mathbf{M}(\mathbf{x}/\varepsilon) \approx \mathbf{M}_0(y_2/\varepsilon, y_3/\varepsilon)$, then

$$\begin{aligned} \sigma_{ij} \approx & [a_{ij11} + a_{ijkl} N_{0k,ly}^{(1)}(y_2/\varepsilon, y_3/\varepsilon)] w_1'(x_1) + \sum_{\alpha=2,3} [-\varepsilon y_\alpha a_{ij11} + a_{ijkl} N_{0k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(x_1) \\ & + [\varepsilon(y_2 a_{ij31} - y_3 a_{ij21}) + a_{ijkl} M_{0k,ly}(y_2/\varepsilon, y_3/\varepsilon)] \Theta'(x_1). \end{aligned}$$

This formula coincides with the corresponding formula of the asymptotic theory for uniform homogeneous beams [34].

Note 1. We considered above the case when the beams and joint are made of the same homogeneous elastic material: $a_{ijkl} = \text{const}$. The case when materials of beams and/or joint are different can be studied by repeating the previous calculations with some modifications. One can use the method of local perturbation to analyze singular inclusions, defects, etc., in beams.

Note 2. If one considers joined beams by using two-dimensional elasticity theory problem, it is sufficient to put $w_3(x_1) = 0$ and $\Theta(x_1) = 0$ in the previous computations.

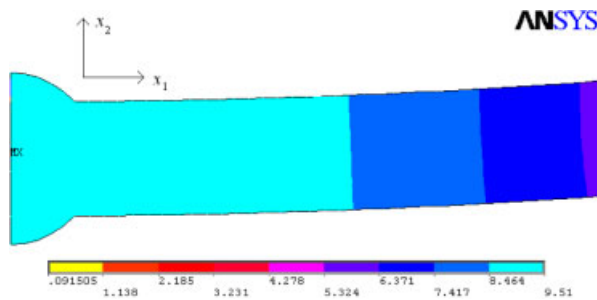


Fig. 2 Numerically computed deformed shape and normal displacement $u_2^\varepsilon(\mathbf{x})$ in two-dimensional problem.

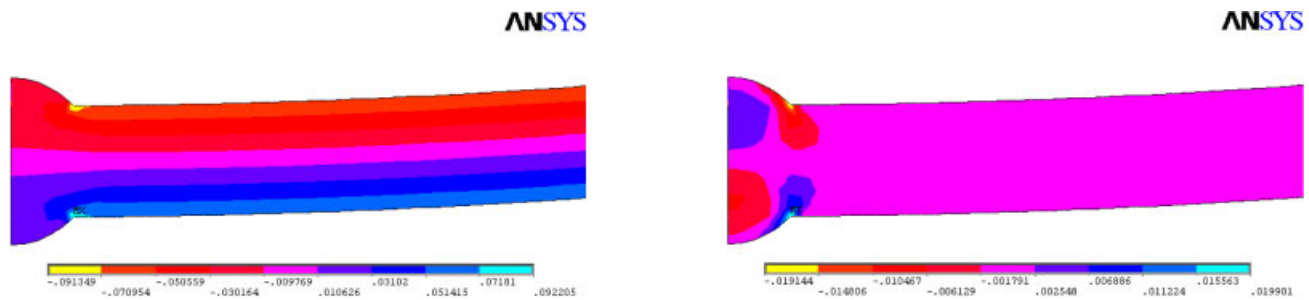


Fig. 3 Numerically computed local stresses $\sigma_{11}(\mathbf{x})$ (left) and $\sigma_{22}(\mathbf{x})$ (right) in two-dimensional problem.

2 Numerical examples

We present numerical results on the problem of bending of joined beams. Both two-dimensional and three-dimensional problems were analyzed. The problems were solved numerically by using ANSYS software [27]. First, we consider two-dimensional numerical results on the problem of bending of two identical homogeneous uniform beams with the joint, which have the shape of circular disk (this problem also can be treated as a problem of cylindrical bending of plates with a circular cylindrical joint). For a reason of symmetry, we can solve the problem for a half of the structure. Fig. 2 displays deformed beam and normal displacements $u_2^\varepsilon(\mathbf{x})$ (Ox_1 -axis is directed along the beam, Ox_2 -axis is normal to the beam). The solution corresponds to the function $w_2(x_1)\mathbf{e}_2 - w_2'(x_1)x_2\mathbf{e}_1 + \varepsilon\mathbf{N}^{(2)}(\mathbf{x}/\varepsilon)w_2'(x_1)$, see Eq. (2) and Note 2 above. The result of the numerical computation supports the assumption that displacement $\mathbf{u}^\varepsilon(\mathbf{x})$ has no jump in the joint. It is in agreement with the engineering theory of elastic frameworks, see, e.g., [19, 26, 27]. Figure 3 (left) displays the local stress $\sigma_{11}(\mathbf{x})$ and Fig. 3 (right) displays the local stress $\sigma_{22}(\mathbf{x})$. It is seen that at the distance about 3-5 diameters of the beam the distribution of the stresses have the form predicted by the engineering theory of beams (i.e. linearity of $\sigma_{11}(\mathbf{x})$ as a function of x_2 and zero value of $\sigma_{22}(\mathbf{x})$).

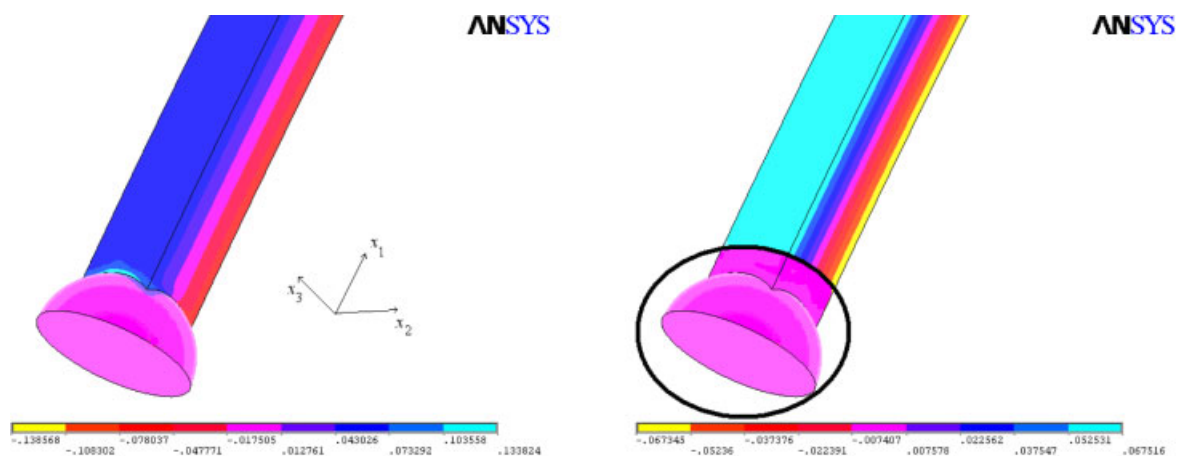


Fig. 4 Numerically computed local stress $\sigma_{11}(\mathbf{x})$ near the joint (left) and in the main part of the beam (right) in three-dimensional problem.

We also analyzed numerically a problem of bending of two identical three-dimensional beams of square cross-section joined with a spherical joint. The system of beams was subjected to bending in Ox_1x_3 -plane. The solution to the problem has the form $w_3(x_1)\mathbf{e}_3 - w'_3(x_1)x_3\mathbf{e}_1 + \varepsilon\mathbf{N}^{(3)}(\mathbf{x}/\varepsilon)w'_3(x_1)$. Since the structure is symmetric, solution may be done for the half of the structure. Figure 4 (left) displays local stress $\sigma_{11}(\mathbf{x})$ in the beam including domain in the vicinity of joint. Since a concentration of stress takes place near the joint, the picture of local stresses outside the joint (in the main part of the beam) is not contrast. More contrast picture can be obtained if remove data in the vicinity of joint (inside the domain marked in Fig. 4 (right) by the bold line) by using an appropriate ANSYS option. Figure 4 (right) displays more contrast distribution of local stress $\sigma_{11}(\mathbf{x})$ in the main part of the beam. It is clear seen the linear distribution of stress $\sigma_{11}(\mathbf{x})$ through the beam.

3 Joined beams of periodic structure

In [15] it was noted that the method of local perturbation can be integrated with the homogenization method for periodic structures, see [1, 5, 6, 13, 23, 32]. In this section, we present the method of local perturbation as applied to joined beams of periodic structure, see Fig. 5.

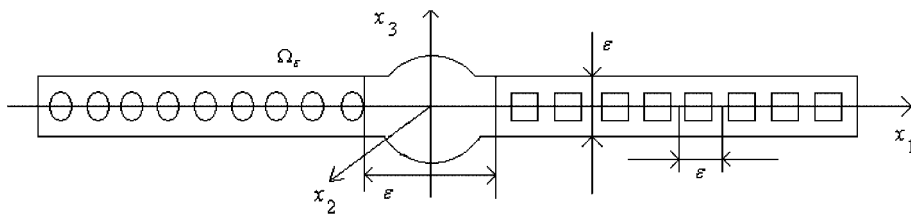


Fig. 5 Two joined beams of periodic structure.

We consider domain Ω_ε , which consists of two thin perforated domains of periodic structure (referred in the sequel “beams of periodic structure”) and joint between them. We assume that all sizes of the joint and all dimensions of the periodicity cells of the beams are proportional to a small parameter ε . We denote period of the left beam in direction Ox_1 by εT_1 and period of the right beam by εT_2 . In the case under consideration $a_{ijkl} = a_{ijkl}(\mathbf{x}/\varepsilon)$ and $\sigma_{ij}n_j = 0$ on the top and bottom boundaries of the beams and holes (for perforated beams like in Fig. 5).

As above, we assume the beams are clamped at $x_1 = \pm 1$ and subjected to the mass force $\mathbf{f}^\varepsilon = (\varepsilon f_1, \varepsilon^2 f_2, \varepsilon^2 f_3)$. The elasticity problem can be written in the form of the minimization problem (1).

In the case under consideration, we propose to search solution $\mathbf{u}^\varepsilon(\mathbf{x})$ in the form

$$\begin{aligned} \mathbf{u}^\varepsilon(\mathbf{x}) = & \varepsilon w_1(x_1)\mathbf{e}_1 + w_2(x_1)\mathbf{e}_2 - w'_2(x_1)x_2\mathbf{e}_1 + w_3(x_1)\mathbf{e}_3 - w'_3(x_1)x_3\mathbf{e}_1 \\ & + (y_2\mathbf{e}_3 - y_3\mathbf{e}_2)\Theta(x_1) + \varepsilon\mathbf{v}(\mathbf{x}/\varepsilon) + \varepsilon\mathbf{v}_p(\mathbf{x}/\varepsilon), \end{aligned} \quad (30)$$

where $\mathbf{v}(\mathbf{y})$ is the function localized near zero and function $\mathbf{v}_p$ is periodic in y_1 with period T_1 in the left beam and with period T_2 in the right beam, see Fig. 5.

Unlike to the previous section, we assume that function $\mathbf{v}(\mathbf{y})$ is not only perturbed but localized near the point $\mathbf{x} = 0$, i.e., $\mathbf{v}(\mathbf{y}) \rightarrow 0$ as $|y_1| \rightarrow \infty$. It is evident that the localization condition is stronger rather than condition of local perturbation. We can stay the localization condition for $\mathbf{v}(\mathbf{y})$ because the function $\mathbf{v}_p(\mathbf{y})$ adsorbs the terms like $\mathbf{N}_0^{(i)}(y_2, y_3)$ ($i = 1, 2, 3$) and $\mathbf{M}_0(y_2, y_3)$. The similar representation of solution can be used for cylindrical beams as alternative to the approach presented in Sect. 2. In other words, we use two correctors: local corrector $\varepsilon\mathbf{v}_p(\mathbf{x}/\varepsilon)$ and periodic corrector $\varepsilon\mathbf{v}_p(\mathbf{x}/\varepsilon)$ of the homogenization theory in the version for beams [20]. Generally, the corrector $\varepsilon\mathbf{v}_p(\mathbf{x}/\varepsilon)$ has different period for $x_1 < 0$ and $x_1 > 0$. It is possible to interpret function $\mathbf{v}_p(\mathbf{y})$ as two functions: $\mathbf{v}_{p1}(\mathbf{y})$ periodic with period T_1 for $x_1 < 0$ and $\mathbf{v}_{p2}(\mathbf{y})$ periodic with period T_2 for $x_1 > 0$.

The derivatives of the displacements (30) are (compare formulas (31)–(33) with formulas (3)–(5))

$$\begin{aligned} u_{k,l}^\varepsilon(\mathbf{x}) = & \varepsilon\delta_{k1}w'_1(x_1)\delta_{l1} + \delta_{k2}w'_2(x_1)\delta_{l1} - \delta_{k1}w'_2(x_1)\delta_{l2} - \varepsilon y_2\delta_{k1}w''_2(x_1)\delta_{l1} \\ & + \delta_{k3}w'_3(x_1)\delta_{l1} - \delta_{k1}w'_3(x_1)\delta_{l3} - \varepsilon y_3\delta_{k1}w''_3(x_1)\delta_{l1} \\ & + (\delta_{l2}\delta_{k3} - \delta_{l3}\delta_{k2})\Theta(x_1) + \varepsilon(y_2\delta_{k3} - y_3\delta_{k2})\Theta'(x_1)\delta_{l1} + v_{k,ly}(\mathbf{y}) + v_{pk,ly}(\mathbf{y}). \end{aligned} \quad (31)$$

The local stresses corresponding to the displacements (31) are

$$\begin{aligned}\sigma_{ij} = & \varepsilon a_{ij11}(\mathbf{y}) w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_{\alpha} a_{ij11}(\mathbf{y}) w_{\alpha}''(x_1) \\ & + \varepsilon (y_2 a_{ij31}(\mathbf{y}) - y_3 a_{ij21}(\mathbf{y})) \Theta'(x_1) + a_{ijkl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{ijkl}(\mathbf{y}) v_{pk,ly}(\mathbf{y}).\end{aligned}\quad (32)$$

The local energy $E(\mathbf{x}, \mathbf{u}^{\varepsilon})$ corresponding to the displacements $\mathbf{u}^{\varepsilon}(\mathbf{x})$ (30) is

$$\begin{aligned}E(\mathbf{x}, \mathbf{u}^{\varepsilon}) = & \frac{1}{2} \sigma_{11} w_1'(x_1) + \frac{1}{2} [-\sigma_{11} \varepsilon \sum_{\alpha=2,3} y_{\alpha} w_{\alpha}''(x_1) \\ & + (\sigma_{31} y_2 - \sigma_{21} y_3) \Theta'(x_1) + \sigma_{kl} v_{k,ly}(\mathbf{y}) + \sigma_{kl} v_{pk,ly}(\mathbf{y})].\end{aligned}\quad (33)$$

As above, we consider the beam made of isotropic material for which $a_{1213} = a_{1312} = a_{1121} = a_{1131} = 0$. For this case, substituting stresses σ_{ij} (32) into expression (33), we obtain

$$\begin{aligned}E(\mathbf{x}, \mathbf{u}^{\varepsilon}) = & \frac{1}{2} \left[a_{1111}(\mathbf{y}) w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_{\alpha} a_{1111}(\mathbf{y}) w_{\alpha}''(x_1) + a_{11kl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{11kl}(\mathbf{y}) v_{pk,ly}(\mathbf{y}) \right] w_1'(x_1) \\ & - \frac{1}{2} \sum_{\beta=2,3} \left[a_{1111}(\mathbf{y}) w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_{\alpha} a_{1111}(\mathbf{y}) w_{\alpha}''(x_1) + a_{11kl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{11kl}(\mathbf{y}) v_{pk,ly}(\mathbf{y}) \right] \varepsilon y_{\beta} w_{\beta}''(x_1) \\ & + \frac{1}{2} (y_2^2 a_{3131}(\mathbf{y}) + y_3^2 a_{2121}(\mathbf{y})) \Theta'^2(x_1) \\ & + \frac{1}{2} [a_{ij11}(\mathbf{y}) w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_{\alpha} a_{ij11}(\mathbf{y}) w_{\alpha}''(x_1) + \varepsilon (y_2 a_{ij31}(\mathbf{y}) - y_3 a_{ij21}(\mathbf{y})) \Theta'(x_1) \\ & + a_{ijkl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{ijkl}(\mathbf{y}) v_{pk,ly}(\mathbf{y})] v_{i,jy}(\mathbf{y}).\end{aligned}\quad (34)$$

As a result, we arrive at the following minimization problem

$$\begin{aligned}L(w_1, w_2, w_3, \Theta, \mathbf{v}, \mathbf{v}_p) = & \int_{\Omega_{\varepsilon}} E(\mathbf{x}, \mathbf{u}^{\varepsilon}) d\mathbf{x} \\ & - \int_{\Omega_{\varepsilon}} \mathbf{f}^{\varepsilon}(x_1) [\varepsilon w_1(x_1) \mathbf{e}_1 + w_2(x_1) \mathbf{e}_2 - w_2'(x_1) x_2 \mathbf{e}_1 + w_3(x_1) \mathbf{e}_3 - w_3'(x_1) x_3 \mathbf{e}_1 \\ & + (y_2 \mathbf{e}_3 - y_3 \mathbf{e}_2) \Theta(x_1) + \varepsilon \mathbf{v}(\mathbf{x}/\varepsilon) + \varepsilon \mathbf{v}_p(\mathbf{x}/\varepsilon)] d\mathbf{x} \rightarrow \min; \\ & w_1, \Theta \in H^1[-1, 1]; \quad w_2, w_3 \in H^2[-1, 1]; \quad \mathbf{v} \in V_{loc}; \quad \mathbf{v}_p \in V_p,\end{aligned}\quad (35)$$

where $E(\mathbf{x}, \mathbf{u}^{\varepsilon})$ is given by expression (34) and $V_p = \{ \mathbf{u} \in V : \mathbf{u}(\mathbf{x}) \text{ is periodic with period } T_1 \text{ for } x_1 < 0 \text{ and with period } T_2 \text{ for } x_1 > 0 \}$.

Functional $L(w_1, w_2, w_3, \Theta, \mathbf{v}, \mathbf{v}_p)$ depends on the six variables: functions $w_1(x_1)$, $w_2(x_1)$, $w_3(x_1)$, $\Theta(x_1)$ of the “slow” variable x_1 and functions $\mathbf{v}(\mathbf{y})$ and $\mathbf{v}_p(\mathbf{y})$ of the “fast” variables $\mathbf{y}$. Euler-Lagrange equations for the functional (35) are the following:

$$\begin{aligned}\delta L_{\mathbf{v}} = & \int_{\Omega_{\varepsilon}} [\varepsilon a_{11kl}(\mathbf{y}) w_1'(x_1) - \varepsilon \sum_{\alpha=2,3} y_{\alpha} a_{1111}(\mathbf{y}) w_{\alpha}''(x_1) \\ & + \frac{1}{2} \varepsilon (y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y})) \Theta'(x_1) + a_{ijkl}(\mathbf{y}) v_{i,jy}(\mathbf{y}) + a_{ijkl}(\mathbf{y}) v_{pi,jy}(\mathbf{y})] \delta v_{k,ly}(\mathbf{y}) d\mathbf{x} \\ & - \int_{\Omega_{\varepsilon}} (\varepsilon^2 f_1(x_1) \delta v_1 + \varepsilon^3 \sum_{\alpha=2,3} f_{\alpha}(x_1) \delta v_{\alpha}) d\mathbf{x} = 0,\end{aligned}\quad (36)$$

$$\begin{aligned}
\delta L_{\mathbf{v}_p} = & \int_{\Omega_\varepsilon} [\varepsilon a_{11kl}(\mathbf{y}) w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y}) w''_\alpha(x_1) \\
& + \frac{1}{2} \varepsilon (y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y})) \Theta'(x_1) + a_{ijkl}(\mathbf{y}) v_{i,jy}(\mathbf{y}) + a_{ijkl}(\mathbf{y}) v_{pi,jy}(\mathbf{y})] \delta v_{pk,ly}(\mathbf{y}) d\mathbf{x} \\
& - \int_{\Omega_\varepsilon} (\varepsilon^2 f_1(x_1) \delta v_{p1} + \varepsilon^3 \sum_{\alpha=2,3} f_\alpha(x_1) \delta v_{p\alpha}^\varepsilon) d\mathbf{x} = 0,
\end{aligned} \quad (37)$$

and

$$\begin{aligned}
\delta L_{w_1} = & \int_{\Omega_\varepsilon} [\varepsilon a_{1111} w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{1111}(\mathbf{y}) w''_\alpha(x_1) \\
& + a_{11kl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{11kl}(\mathbf{y}) v_{pk,ly}(\mathbf{y})] \varepsilon \delta w'_1(x_1) d\mathbf{x} - \int_{\Omega_\varepsilon} \varepsilon^2 f_1(x_1) \delta w_1(x_1) d\mathbf{x} = 0,
\end{aligned} \quad (38)$$

$$\begin{aligned}
\delta L_{w_\alpha} = & \int_{\Omega_\varepsilon} [\varepsilon a_{1111}(\mathbf{y}) w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{1111}(\mathbf{y}) w''_\alpha(x_1) \\
& + a_{11kl}(\mathbf{y}) v_{k,ly}(\mathbf{y}) + a_{11kl}(\mathbf{y}) v_{pk,ly}(\mathbf{y})] \varepsilon y_\alpha \delta w''_\alpha(x_1) d\mathbf{x} \\
& - \int_{\Omega_\varepsilon} [\varepsilon^2 f_\alpha(x_1) \delta w_2(x_1) - \varepsilon^2 f_1(x_1) \delta w'_\alpha(x_1) y_2] d\mathbf{x} = 0 \quad (\alpha = 2, 3),
\end{aligned} \quad (39)$$

$$\begin{aligned}
\delta L_\Theta = & \int_{\Omega_\varepsilon} [\varepsilon^2 (y_2^2 a_{3131}(\mathbf{y}) + y_3^2 a_{2121}(\mathbf{y})) \Theta'(x_1) + \frac{1}{2} \varepsilon (y_2 a_{ij31}(\mathbf{y}) - y_3 a_{ij21}(\mathbf{y})) v_{i,jy}(\mathbf{y}) \\
& + \frac{1}{2} \varepsilon (y_2 a_{ij31}(\mathbf{y}) - y_3 a_{ij21}(\mathbf{y})) v_{pi,jy}(\mathbf{y})] \delta \Theta'(x_1) d\mathbf{x} - \int_{\Omega_\varepsilon} \varepsilon^3 [f_3(x_1) y_2 - f_2(x_1) y_3] \Theta(x_1) d\mathbf{x}.
\end{aligned} \quad (40)$$

In Eqs. (36) and (37), the leading terms have orders of ε . We omit the terms of order higher than ε in Eqs. (36) and (37).

It is convenient to introduce function $\mathbf{W}(\mathbf{x}/\varepsilon) = \mathbf{v}(\mathbf{x}/\varepsilon) + \mathbf{v}_p(\mathbf{x}/\varepsilon)$. This is a sum of function $\mathbf{v}_p(\mathbf{x}/\varepsilon) \in V_p$ and function $\mathbf{v}(\mathbf{y})$ localized near the point $\mathbf{y} = 0$. As a result, function $\mathbf{W}(\mathbf{x}/\varepsilon)$ tends to function periodic in y_1 with the period T_1 as $y_1 \rightarrow -\infty$ and period T_2 as $y_1 \rightarrow \infty$. From (36) and (37) we obtain the following variation equation with respect to the function $\mathbf{W}(\mathbf{x}/\varepsilon)$:

$$\begin{aligned}
& \int_{\Omega_\varepsilon} [\varepsilon a_{11kl}(\mathbf{y}) w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y}) w''_\alpha(x_1) + \frac{1}{2} \varepsilon (y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y})) \Theta'(x_1) \\
& + a_{ijkl}(\mathbf{y}) W_{i,jy}(\mathbf{y})] \delta W_{k,ly}(\mathbf{y}) d\mathbf{x} = 0.
\end{aligned} \quad (41)$$

Integrating Eq. (41) by parts, we obtain the following boundary value problem

$$\begin{aligned}
& [a_{ijkl}(\mathbf{y}) W_{i,jy} + \varepsilon a_{11kl}(\mathbf{y}) w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y}) w''_\alpha(x_1) \\
& + \frac{1}{2} \varepsilon (y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y})) \Theta'(x_1)]_{,ly} = 0 \quad \text{in } \Omega_\varepsilon, \\
& [a_{ijkl}(\mathbf{y}) W_{i,jy} + \varepsilon a_{11kl}(\mathbf{y}) w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y}) w''_\alpha(x_1) \\
& + \frac{1}{2} \varepsilon (y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y})) \Theta'(x_1)] n_l = 0 \quad \text{on the lateral side of } \Omega_\varepsilon,
\end{aligned} \quad (42)$$

$\mathbf{W}(\mathbf{x}/\varepsilon)$ tends to a function periodic with period εT_1 as $x_1 \rightarrow -\infty$ and period εT_2 as $x_1 \rightarrow \infty$.

The function $\mathbf{v}_p(\mathbf{y})$ in the main parts of the beams is determined from the problem

$$\begin{aligned}
 & [a_{ijkl}(\mathbf{y})v_{pi,jy}(\mathbf{y}) + a_{11kl}(\mathbf{y})w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y})w''_\alpha(x_1) \\
 & + \frac{1}{2}\varepsilon(y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y}))\Theta'(x_1)]_{,ly} = 0 \quad \text{in } \Omega_\varepsilon, \\
 & [a_{ijkl}(\mathbf{y})v_{pi,jy}(\mathbf{y}) + a_{11kl}(\mathbf{y})w'_1(x_1) - \varepsilon \sum_{\alpha=2,3} y_\alpha a_{11kl}(\mathbf{y})w''_\alpha(x_1) \\
 & + \frac{1}{2}\varepsilon(y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y}))\Theta'(x_1)]n_l = 0 \quad \text{on the lateral side of } \Omega_\varepsilon, \\
 & \mathbf{v}_p(\mathbf{x}/\varepsilon) \text{ is periodic with period } \varepsilon T_1 \text{ as } x_1 < 0 \text{ and period } \varepsilon T_2 \text{ as } x_1 > 0,
 \end{aligned} \tag{43}$$

which follows from Eqs. (42) and condition that $\mathbf{v}(\mathbf{y})$ is localized near the point $\mathbf{x} = 0$. Separating fast and slow variables in Eqs. (43), we can write solution to Eq. (42) in the form

$$\mathbf{W}(\mathbf{x}/\varepsilon) = \varepsilon \mathbf{N}^{(1)}(\mathbf{x}/\varepsilon)w'_1(x_1) + \varepsilon \sum_{\alpha=2,3} \mathbf{N}^{(\alpha)}(\mathbf{x}/\varepsilon)w''_\alpha(x_1) + \varepsilon \mathbf{M}(\mathbf{x}/\varepsilon)\Theta'(x_1), \tag{44}$$

where $\mathbf{N}^{(1)}(\mathbf{y})$ is solution to the following boundary value problem in domain $\Omega_\infty(\varepsilon) = \frac{1}{\varepsilon}\Omega_\varepsilon = \{\mathbf{y} = \frac{1}{\varepsilon}\mathbf{x} : \mathbf{x} \in \Omega_\varepsilon\}$:

$$\begin{aligned}
 & [a_{ijkl}(\mathbf{y})N_{i,jy}^{(1)}(\mathbf{y}) + a_{11kl}(\mathbf{y})]_{,ly} = 0 \quad \text{in } \Omega_\infty(\varepsilon), \\
 & [a_{ijkl}(\mathbf{y})N_{i,jy}^{(1)}(\mathbf{y}) + a_{11kl}(\mathbf{y})]n_l = 0 \quad \text{on lateral side of } \Omega_\infty(\varepsilon), \\
 & \mathbf{N}^{(1)}(\mathbf{y}) \text{ tends to a function periodic with period } T_1 \text{ as } y_1 \rightarrow -\infty \text{ and period } T_2 \text{ as } y_1 \rightarrow \infty.
 \end{aligned} \tag{45}$$

Function $\mathbf{N}^{(\alpha)}(\mathbf{y})$ ($\alpha = 2, 3$) is solution to the problem

$$\begin{aligned}
 & [a_{ijkl}(\mathbf{y})N_{i,jy}^{(\alpha)}(\mathbf{y}) - y_\alpha a_{11kl}(\mathbf{y})]_{,ly} = 0 \quad \text{in } \Omega_\infty(\varepsilon), \\
 & [a_{ijkl}(\mathbf{y})N_{i,jy}^{(\alpha)}(\mathbf{y}) - y_\alpha a_{11kl}(\mathbf{y})]n_l = 0 \quad \text{on lateral side of } \Omega_\infty(\varepsilon), \\
 & \mathbf{N}^{(\alpha)}(\mathbf{y}) \text{ tends to a function periodic with period } T_1 \text{ as } y_1 \rightarrow -\infty \text{ and period } T_2 \text{ as } y_1 \rightarrow \infty.
 \end{aligned} \tag{46}$$

Function $\mathbf{M}(\mathbf{y})$ is solution to the problem

$$\begin{aligned}
 & [a_{ijkl}(\mathbf{y})M_{i,jy}(\mathbf{y}) - \frac{1}{2}(y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y}))]_{,ly} = 0 \quad \text{in } \Omega_\infty(\varepsilon), \\
 & [a_{ijkl}(\mathbf{y})M_{i,jy}(\mathbf{y}) - \frac{1}{2}(y_2 a_{kl31}(\mathbf{y}) - y_3 a_{kl21}(\mathbf{y}))]n_l = 0 \quad \text{on lateral side of } \Omega_\infty(\varepsilon), \\
 & \mathbf{M}(\mathbf{y}) \text{ tends to a function periodic with period } T_1 \text{ as } y_1 \rightarrow -\infty \text{ and period } T_2 \text{ as } y_1 \rightarrow \infty.
 \end{aligned} \tag{47}$$

For the function $\mathbf{v}_p(\mathbf{y})$ we have the representation similar to Ansatz (44):

$$\mathbf{v}_p(\mathbf{x}/\varepsilon) = \varepsilon \mathbf{N}_p^{(1)}(\mathbf{x}/\varepsilon)w'_1(x_1) + \varepsilon \sum_{\alpha=2,3} \mathbf{N}_p^{(\alpha)}(\mathbf{x}/\varepsilon)w''_\alpha(x_1) + \varepsilon \mathbf{M}_p(\mathbf{x}/\varepsilon)\Theta'(x_1), \tag{48}$$

where $\mathbf{N}_p^{(i)}(\mathbf{y})$ ($i = 1, 2, 3$) and $\mathbf{M}_p(\mathbf{y})$ are solutions to the problems obtained from Eqs. (45) and (47) by the changing the condition “tends to a periodic function with the period T_1 as $y_1 \rightarrow -\infty$ and period T_2 as $y_1 \rightarrow \infty$ ” for the condition “is periodic with the period T_1 as $y_1 < 0$ and period T_2 as $y_1 > 0$ ”. It means that functions $\mathbf{N}_p^{(i)}(\mathbf{y})$ ($i = 1, 2, 3$) and $\mathbf{M}_p(\mathbf{y})$ are solutions to the cell problems of the homogenization theory in the version for beams (see, e.g., [20]), on the semi axes $y_1 < 0$ and $y_1 > 0$. To the best knowledge of the authors, there exist no explicit solutions to the cell problems for beams. Functions $\mathbf{N}_p^{(i)}(\mathbf{y})$ ($i = 1, 2, 3$) and $\mathbf{M}_p(\mathbf{y})$ are computed numerically.

Substituting Ansatz (45) into Eq. (38) and dividing by ε^{-4} (arguments for dividing by ε^{-4} are given after (17)), we have after the collecting of terms

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ y_2 [a_{1111}(\mathbf{y}) + y_2 a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) + \sum_{\alpha=2,3} [y_\alpha a_{1111}(\mathbf{x}/\varepsilon) + a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(x_1) \right. \\ & \left. + y_2 a_{11kl}(\mathbf{x}/\varepsilon) M_{k,ly}(\mathbf{x}/\varepsilon) \Theta(x_1) \right\} \delta w_1'(x_1) d\mathbf{x} - \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} f_1(x_1) \delta w_1(x_1) d\mathbf{x} = 0. \end{aligned} \quad (49)$$

Substituting Ansatz (44) into Eq. (38) for $\alpha = 2$ and dividing by ε^{-4} , we have

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ y_2 [a_{1111}(\mathbf{x}/\varepsilon) + a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) \right. \\ & + [-y_2^2 a_{1111}(\mathbf{x}/\varepsilon) + y_2 a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(2)}(\mathbf{x}/\varepsilon)] w_2''(x_1) \\ & + [y_2 y_3 a_{1111}(\mathbf{x}/\varepsilon) + y_2 a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(3)}(\mathbf{x}/\varepsilon)] w_3''(x_1) \\ & \left. + y_2 a_{11kl}(\mathbf{x}/\varepsilon) M_{k,ly}(\mathbf{x}/\varepsilon) \Theta'(x_1) \right\} \delta w_2''(x_1) d\mathbf{x} \\ & - \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} [f_2(x_1) \delta w_2(x_1) - f_1(x_1) y_2 \delta w_2'(x_1)] d\mathbf{x} = 0. \end{aligned} \quad (50)$$

Substituting Ansatz (44) into Eq. (39) for $\alpha = 3$ and dividing by ε^{-4} , we have

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ y_3 [a_{1111}(\mathbf{x}/\varepsilon)] + a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon) \right\} w_1'(x_1) \\ & + [-y_2 y_3 a_{1111}(\mathbf{x}/\varepsilon) + \frac{1}{2} y_2 a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(2)}(\mathbf{x}/\varepsilon)] w_2''(x_1) \\ & + [-y_3^2 a_{1111}(\mathbf{x}/\varepsilon) + y_3 a_{11kl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(3)}(\mathbf{x}/\varepsilon)] w_3''(x_1) + y_3 a_{11kl}(\mathbf{x}/\varepsilon) M_{k,ly}(\mathbf{x}/\varepsilon) \Theta'(x_1) \left\} \delta w_3''(x_1) d\mathbf{x} \right. \\ & \left. - \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} [f_3(x_1) \delta w_3(x_1) - f_1(x_1) y_3 \delta w_3'(x_1)] d\mathbf{x} = 0. \right. \end{aligned} \quad (51)$$

Substituting Ansatz (44) into Eq. (39) and dividing by ε^{-4} , we obtain

$$\begin{aligned} & \frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} \left\{ [(y_2^2 a_{3131}(\mathbf{x}/\varepsilon) + y_3^2 a_{2121}(\mathbf{x}/\varepsilon)) + \frac{1}{2} (y_2 a_{ij31}(\mathbf{x}/\varepsilon) \right. \\ & - y_3 a_{ij21}(\mathbf{x}/\varepsilon)) M_{i,jy}(\mathbf{x}/\varepsilon)] \Theta'(x_1) + \frac{1}{2} (y_2 a_{ij31}(\mathbf{x}/\varepsilon) - y_3 a_{ij21}(\mathbf{x}/\varepsilon)) N_{i,jy}^{(1)}(\mathbf{x}/\varepsilon) w_1'(x_1) \\ & \left. + \frac{1}{2} \sum_{\alpha=2,3} (y_2 a_{ij31}(\mathbf{x}/\varepsilon) - y_3 a_{ij21}(\mathbf{x}/\varepsilon)) N_{i,jy}^{(\alpha)}(\mathbf{x}/\varepsilon) w_\alpha''(x_1) \right\} \delta \Theta'(x_1) d\mathbf{x} = 0. \end{aligned} \quad (52)$$

We consider the perturbations $\mathbf{n}^{(i)}(\mathbf{y}) = \mathbf{N}^{(i)}(\mathbf{y}) - \mathbf{N}_p^{(i)}(\mathbf{y})$ and $\mathbf{m}(\mathbf{y}) = \mathbf{M}(\mathbf{y}) - \mathbf{M}_p(\mathbf{y})$. Under the condition $\nabla \mathbf{n}^{(i)}(\mathbf{y}), \nabla \mathbf{m}(\mathbf{y}) \in L(\Omega_\infty)$, integrals $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x}$, $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} a_{11kl} m_{k,ly}(\mathbf{x}/\varepsilon) d\mathbf{x}$, $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} y_\beta a_{11kl} n_{k,ly}^{(i)}(\mathbf{x}/\varepsilon) d\mathbf{x}$, and $\frac{1}{\varepsilon^2} \int_{\Omega_\varepsilon} y_\beta a_{11kl} m_{k,ly}(\mathbf{x}/\varepsilon) d\mathbf{x}$ ($i = 1, 2, 3$; $\beta = 2, 3$) tend to zero, as $\varepsilon \rightarrow 0$. It can be demonstrated as above, see Eqs. (23) and (24). Thus, in the limit, as $\varepsilon \rightarrow 0$, we can save in Eqs. (49)–(52) only the periodic components $\mathbf{N}_p^{(i)}(\mathbf{y})$ ($i = 1, 2, 3$) and $\mathbf{M}_p(\mathbf{y})$ of the functions $\mathbf{N}^{(i)}(\mathbf{y})$ and $\mathbf{M}(\mathbf{y})$ (i.e., change in Eqs. (49)–(52) $\mathbf{N}^{(i)}(\mathbf{y})$ and $\mathbf{M}(\mathbf{y})$ for $\mathbf{N}_p^{(i)}(\mathbf{y})$ and $\mathbf{M}_p(\mathbf{y})$). After we have made this changing, the variation Eqs. (49)–(52) coincide with the corresponding equations in the homogenization theory in the version for beams [20]. The limit equations of the homogenization theory of beams are known. They have

the form (25) and (26). The coefficients of the equations are determined with accordance to the homogenization theory for beams and $F_i(x_1) = \frac{1}{T_i} \int_{Y_i} f_i(x_1) d\mathbf{y}$, $m_\alpha(x_1) = \frac{1}{T_i} \int_{Y_i} f_1(x_1) y_\alpha d\mathbf{y}$, where Y_i ($i = 1, 2$) means the periodicity cell of the left ($i = 1$) or the right ($i = 2$) beams. The coefficients in the limit problem (25) and (26) do not depend on the geometry or material of the joint. They depend on the geometry or material characteristics of the periodicity of the periodicity cells. We conclude that the joint of normal type does not influence the property of the joined beams on global level.

It is not clear if it is possible the vanishing coupled coefficients $D_{1\alpha}$, $D_{\alpha 1}$, F_i , J_1 and I , simultaneously. For plate similar problem is investigated more carefully [22].

For the reason presented before the Eq. (29), the local stresses in the vicinity of joint in accordance with Eqs. (32) and (44) are

$$\begin{aligned} \sigma_{ij} \approx & [a_{ij11}(\mathbf{x}/\varepsilon) + a_{ijkl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(0) \\ & + \sum_{\alpha=2,3} [-\varepsilon y_\alpha a_{ij11}(\mathbf{x}/\varepsilon) + a_{ijkl}(\mathbf{x}/\varepsilon) N_{k,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(0) \\ & + [\varepsilon(y_2 a_{ij31}(\mathbf{x}/\varepsilon) - y_3 a_{ij21}(\mathbf{x}/\varepsilon)) + a_{ijkl}(\mathbf{x}/\varepsilon) M_{k,ly}(\mathbf{x}/\varepsilon)] \Theta'(0). \end{aligned} \quad (53)$$

Since in the main part of the beams $\mathbf{N}^{(i)}(\mathbf{x}/\varepsilon) \approx \mathbf{N}_p^{(i)}(\mathbf{x}/\varepsilon)$ and $\mathbf{M}(\mathbf{x}/\varepsilon) \approx \mathbf{M}_p(\mathbf{x}/\varepsilon)$, then the local stresses in the main parts of the beams are

$$\begin{aligned} \sigma_{ij} \approx & [a_{ij11}(\mathbf{x}/\varepsilon) + a_{ijkl}(\mathbf{x}/\varepsilon) N_{pk,ly}^{(1)}(\mathbf{x}/\varepsilon)] w_1'(x_1) \\ & + \sum_{\alpha=2,3} [-\varepsilon y_\alpha a_{ij11}(\mathbf{x}/\varepsilon) + a_{ijkl}(\mathbf{x}/\varepsilon) N_{pk,ly}^{(\alpha)}(\mathbf{x}/\varepsilon)] w_\alpha''(x_1) \\ & + [\varepsilon(y_2 a_{ij31}(\mathbf{x}/\varepsilon) - y_3 a_{ij21}(\mathbf{x}/\varepsilon)) + a_{ijkl}(\mathbf{x}/\varepsilon) M_{pk,ly}(\mathbf{x}/\varepsilon)] \Theta'(x_1). \end{aligned} \quad (54)$$

It is seen from the formula (53) that the local stresses in the vicinity of joint are determined by the functions $\mathbf{N}^{(i)}(\mathbf{x})$ ($i = 1, 2, 3$) and $\mathbf{M}(\mathbf{x})$, which essentially depend on the geometry and material of the joint (see problems (44) and (45), which introduce the mentioned functions). In the main parts of the beams, the local stresses are determined by the functions $\mathbf{N}_p^{(i)}(\mathbf{x})$ ($i = 1, 2, 3$) and $\mathbf{M}_p(\mathbf{x})$ (see formula (54)), which are depend on the geometry and material properties of the periodicity cells.

4 Numerical examples

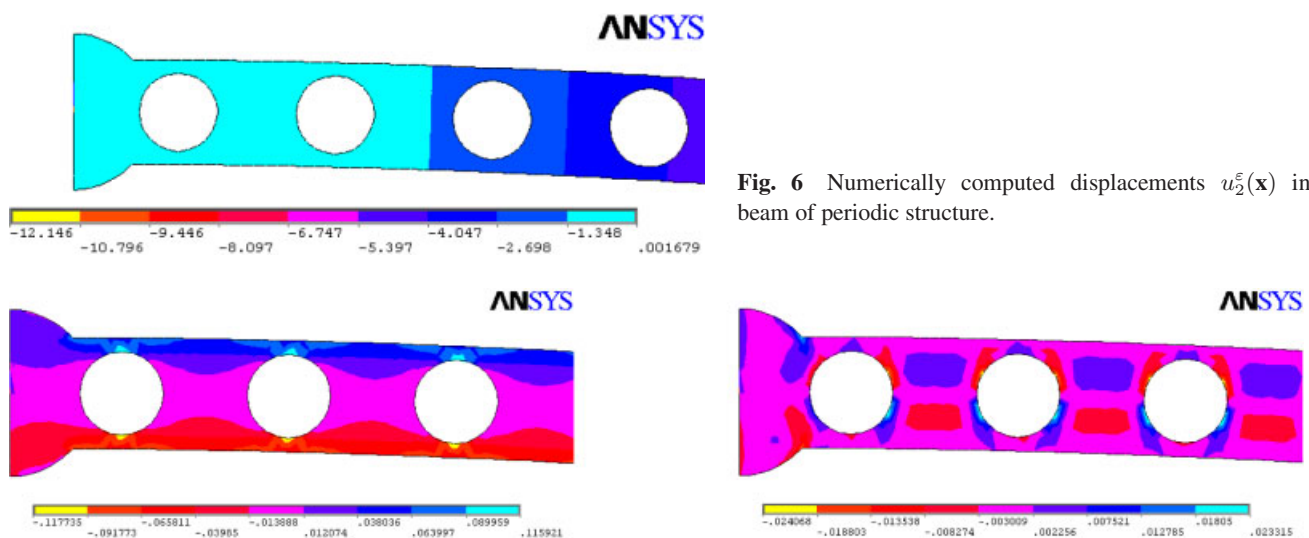


Fig. 7 Numerically computed local stresses $\sigma_{11}(\mathbf{x})$ (left) and $\sigma_{22}(\mathbf{x})$ (right) in beam of periodic structure.

We present results of the numerical computations for two-dimensional elasticity problems for joined beams of periodic structure. We consider two-dimensional problem about bending of two identical beams with periodical system of circular

holes with joint, which has a form of circular disk. As a result of the symmetry of the problem, we can solve the problem for a half of the structure. Figure 6 displays numerically computed deformed shape and displacements $u_2^\varepsilon(\mathbf{x})$ of the beam. It is seen that displacement has no jump in the joint.

Figure 7 (left) displays numerically computed local stress $\sigma_{11}(\mathbf{x})$. Figure 7 (right) displays numerically computed local stress $\sigma_{22}(\mathbf{x})$. It is seen that at the distance about 3-5 diameters of the beam the distribution of the stresses became periodic.

5 Concluding remarks

The local perturbation method (presented in this paper as applied to joined elastic rods and in [15] as applied to conductivity problems) is based on the idea of two-scale decomposition of solution. Two scale approach leads to problems in “fast” variables, which correspond to consideration of joint (more generally, a geometrical or material inhomogeneity) in uniform global field. Similar situation arises in the homogenization theory. But there exists an essential difference. The homogenization theory usually considers bodies of periodic structure. As a result, one arrives at the cell problems [5, 6, 13]. In our case, we arrive at problems with the conditions of local perturbation / localization of solution near the joint (see, e.g., problems (14)–(16)). We conclude that the universal idea is the consideration of inhomogeneity subjected to a uniform global field. The idea of periodicity cell problem (extensively and productively utilized in the homogenization theory as well as in the theory of composite materials and complex structures) and idea of local perturbation (realized first, to the best knowledge of the author, in [15]) are realization of this universal idea.

Application of the local perturbation method to joined elastic beams with a joint of normal type leads to strong decomposition of the original elasticity theory problem into problem about deformation of a system of joined one-dimensional beams and three-dimensional (two-dimensional) elasticity problem about joint in beams subjected to uniform tension, bending or torsion. If there are several joints, the elasticity theory problems for joints are separated (i.e. the problems are solved for every particular joint independently) if joints are placed sufficiently far one from others.

If one has the collection of solutions of problems about deformation of joints subjected to the basic uniform fields: macroscopically uniform tension, bending and torsion (for frameworks the set of basic uniform field is more complex), one can compute local stresses in the following stages: 1) solve problem about deformation of a system of joined one-dimensional beams, 2) compute tension, bending and torsion in one-dimensional beams, 3) form a linear combinations of the stress concentration tensors and tension, bending and torsion determined from solution of one-dimensional problem. The stresses in the main part of the beams can be computed in accordance with the asymptotic theory, see, e.g., [20, 34].

Although solutions of three-dimensional (or two-dimensional) problems about the deformation of joints must be done in “infinite” domain, due to the strong localization of the perturbations of local fields near the joint, it is possible solve these problems in a not large domain. Thus, solutions can be obtained with standard FEM software and the method can be used in engineering computations.

The local perturbations method looks like a prospective tool for analysis of a wide range of engineering problems. The first actual problem is the modification of the local perturbations methods for basic types of engineering structures contained joints. We have the following types of “elementary joints”: solid–solid joint, solid–plate joint, solid–beam joint, plate–plate joint, plate–beam joint, beam–beam joint. In joined domain both scalar or vectorial problem can be formulated. Now, the local perturbations methods are developed for beam–beam joint for scalar problem in [15] and vectorial elasticity theory problem in this paper.

Real engineering structures usually are formed of numerous structural elements assembled by joints of various types. Usually, more than two structural elements are connected by one joint (so called multi-joint). In [15] analysis of multi-joint was presented for scalar problem. Modification of the local perturbations method for analysis of elastic multi-joints looks like a practically important problem.

Another prospective direction of application of the local perturbation method is analysis of singular defects and inclusions in uniform beams. Accounting of a singular inclusion in a uniform beam leads to the representation of the elastic constant tensor in the form $a_{ijkl} + b_{ijkl}(\mathbf{x}/\varepsilon)$, where a_{ijkl} is elastic constants of the “guest” material (it is a constant tensor) and $b_{ijkl}(\mathbf{x}/\varepsilon)$ is elastic constants perturbation (it is localized near the inclusion). It is expected that the local perturbation method will work in this case. The local perturbations method has potential for analysis of semi-classical and non-classical problems, such as joints, inhomogeneities, defects, contacts, etc., in discrete and molecular structures.

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6 Appendix. Solutions of cellular problems for homogeneous cylindrical beams

Solutions $\mathbf{N}_0^{(i)}(y_2, y_3)$ ($i = 1, 2, 3$) and $\mathbf{M}_0(y_2, y_3)$ to cell problems for beam play an important role in the developed theory. Although the cell problems for beam usually cannot be solved in explicit form, solutions $\mathbf{N}_0^{(i)}(y_2, y_3)$ ($i = 1, 2, 3$) may be constructed in explicit form for cylindrical beam [24]. We present these solutions below. Cellular problems for cylindrical beam under the condition

$$a_{1\beta\gamma\delta} = 0 \quad (\beta, \gamma, \delta = 2, 3), \quad (\text{A1})$$

are the following (see for details [20, 34]): problem corresponding to axial tension of beam

$$\begin{aligned} [a_{\alpha\beta\gamma\delta} N_{0\gamma,\delta y}^{(1)}(y_2, y_3) + a_{11\alpha\beta}]_{,\beta y} &= 0 \quad \text{in } S, \\ [a_{\alpha\beta\gamma\delta} N_{0\gamma,\delta y}^{(1)}(y_2, y_3) + a_{11\alpha\beta}] n_\beta &= 0 \quad \text{on lateral side of } S, \end{aligned} \quad (\text{A2})$$

problem corresponding to bending of beam

$$\begin{aligned} [a_{\alpha\beta\gamma\delta} N_{0\gamma,\delta y}^{(\rho)}(y_2, y_3) - y_\rho a_{11\alpha\beta}]_{,\beta y} &= 0 \quad \text{in } S, \\ [a_{\alpha\beta\gamma\delta} N_{0\gamma,\delta y}^{(\rho)}(y_2, y_3) - y_\rho a_{11\alpha\beta}] n_\beta &= 0 \quad \text{on lateral side of } S, \end{aligned} \quad (\text{A3})$$

problem corresponding to torsion of beam

$$\begin{aligned} [a_{\alpha\beta\gamma\delta} M_{0i,jy}(y_2, y_3) - \frac{1}{2}(y_2 a_{\alpha\beta 31} - y_3 a_{\alpha\beta 21})]_{,\beta y} &= 0 \quad \text{in } S, \\ [a_{\alpha\beta\gamma\delta} M_{0i,jy}(y_2, y_3) - \frac{1}{2}(y_2 a_{\alpha\beta 31} - y_3 a_{\alpha\beta 21})] n_\beta &= 0 \quad \text{on lateral side of } S. \end{aligned} \quad (\text{A4})$$

In (A2) and (A3) first components of solutions are zero: $N_{01}^{(1)} = 0$, $N_{01}^{(\rho)} = 0$ ($\rho = 2, 3$), $M_{01} = 0$ and other components (the second and the third ones) depend on the variables y_2 and y_3 .

Condition (A1) is satisfied for isotropic material, for which elastic tensor has the form

$$a_{ijkl} = \frac{E\nu}{(1+\nu)(1-2\nu)} \delta_{ij} \delta_{kl} + \frac{E}{2(1+\nu)} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (\text{A5})$$

Problem (A4) coincides with the problem of classical (Saint-Venant) theory of torsion for homogeneous cylindrical beams.

Problems (A2) and (A3) may be solved in explicit form. Solution to (A2) is

$$N_{0\beta}^{(1)}(y_2, y_3) = -\nu y_\beta. \quad (\text{A6})$$

One can verify that $a_{\alpha\beta\gamma\delta} N_{0\gamma,\delta y}^{(1)}(y_2, y_3) + a_{11\alpha\beta} = 0$ by substituting (A5) and (A6) into this expression.

Solution to (A3) is

$$\begin{aligned} N_{02}^{(2)}(y_2, y_3) &= \nu \frac{y_2^2}{2} - \nu \frac{y_3^2}{2}, & N_{03}^{(2)}(y_2, y_3) &= \nu y_2 y_3, \\ N_{02}^{(3)}(y_2, y_3) &= \nu \frac{y_3^2}{2} - \nu \frac{y_2^2}{2}, & N_{03}^{(3)}(y_2, y_3) &= \nu y_2 y_3. \end{aligned} \quad (\text{A7})$$

One can verify that

$$a_{\alpha\beta\gamma\delta l} N_{0\gamma,\delta y}^{(\rho)}(y_2, y_3) - y_\rho a_{11\alpha\beta} = 0$$

by substituting (A5) and (A3) into this expression.

References

- [1] I. V. Andrianov, J. Awrejcewicz, and L. I. Manevitch, *Asymptotical Mechanics of Thin-Walled Structures: A Handbook* (Springer-Verlag, Berlin, 2004).
- [2] B. D. Annin, A. L. Kalamkarov, A. G. Kolpakov, and V. Z. Parton, *Computation and Design of Composite Material and Structural Elements* (in Russian) (Nauka, Novosibirsk, 1993).
- [3] J. Awrejcewicz, I. V. Andrianov, and L. I. Manevitch, *Asymptotic Approach in Nonlinear Dynamics: New Trends and Applications* (Springer-Verlag, Berlin, 1998).
- [4] J. Awrejcewicz, V. A. Krysko, and A. F. Vakakis, *Nonlinear Dynamics of Continuous Elastic Systems* (Springer-Verlag, Berlin, 2004).
- [5] N. S. Bakhvalov and G. P. Panasenko, *Homogenisation: Averaging Processes in Periodic Media: Mathematical Problems in the Mechanics of Composite Materials* (Kluwer Academic Publ., Dordrecht, 1989).
- [6] A. Bensoussan, J.-L. Lions, and G. Papanicolaou, *Asymptotic Analysis for Periodic Structures* (North Holland Publ., Amsterdam, 1978).
- [7] J. Bickford, *An Introduction to the Design and Behavior of Bolted Joints*, 3rd ed. (Marcel Dekker, New York, 1995).
- [8] A. Blake, *Design of Mechanical Joints* (CRC Press, Boca Raton, FL, 1985).
- [9] D. Blanchard, A. Gaudiello, and G. Griso, Junction of a periodic family of elastic rods with a 3D plate, Part I, *J. Math. Pure Appl.* **88**(1), 133 (2007).
- [10] E. Cabib, L. Freddi, A. Morassi, and D. Percivale, Thin notched beams, *J. Elast.* **64**, 157178 (2001).
- [11] W. F. Chen and E. M. Lui, *Handbook of Structural Engineering* (CRC Press, Boca Raton, FL, 2005).
- [12] P. G. Ciarlet, *Plates and Junctions in Elastic Multi-Structures: An Asymptotic Analysis* (Springer-Verlag, Berlin, 1991).
- [13] D. Cioranescu and P. Donato, *An Introduction to Homogenization* (Oxford University Press, New York, 1999).
- [14] J. A. Collins, G. H. Staab, and H. R. Busby, *Mechanical Design of Machine Elements and Machines* (Wiley, Chichester, 2002).
- [15] A. Gaudiello and A. G. Kolpakov, Influence of nondegenerated joint on the global and local behavior of joined rods, *Int. J. Eng. Sci.* **49**, 295309 (2011).
- [16] A. Gaudiello, R. Monneau, J. Mossino, J. Murat, and A. Sili, Junction of elastic plates and beams, *ESAIM Control Optim. Calc. Var.* **13**(3), 419457 (2007).
- [17] A. L. Gol'denveizer, Algorithms of the asymptotic construction of linear two-dimensional thin shell theory and the St. Venant principle, *J. Appl. Math. Mech.* **58**(6), 10391050, (1994).
- [18] A. L. Gol'denveizer and J. D. Kaplunov, Dynamic boundary layer in problem of vibration of shells, *Mech. Solids* **23**(4), 146155 (1988).
- [19] M. Ivanyi and C. C. Baniotopoulos, *Semi-rigid Joints in Structural Steelwork* (Springer-Verlag, Berlin, 2000).
- [20] A. L. Kalamkarov and A. G. Kolpakov, *Analysis, Design and Optimization of Composite Structures* (Wiley, Chichester, 1997).
- [21] A. Klarbring and A. B. Movchan, Asymptotic modelling of adhesive joints, *Mech. Mater.* **28**(1-4), 137145 (1998).
- [22] A. A. Kolpakov, Design of laminated plate possessing the required stiffnesses using the minimal number of materials and layers, *J. Elast.* **86**, 245261 (2007).
- [23] A. A. Kolpakov and A. G. Kolpakov, *Capacity and Transport in Contrast Composite Structures: Asymptotic Analysis and Applications* (CRC Press, Boca Raton, FL, 2010).
- [24] A. G. Kolpakov, The stiffnesses of elastic cylindrical beams, *J. Appl. Math. Mech.* **58**, 291301 (1994).
- [25] A. G. Kolpakov, *Stressed Composite Structures: Homogenized Models for Thin-Walled Nonhomogeneous Structures with Initial Stresses* (Springer-Verlag, Berlin, 2004).
- [26] S. Krenk, *Mechanics and Analysis of Beams, Columns and Cables: A Modern Introduction to the Classic Theories* (Springer-Verlag, Berlin, 2001).
- [27] R. W. Messler, *Integral Mechanical Attachment* (Butterworth-Heinemann, Oxford, 2006).
- [28] M. Meyyappan, *Carbon Nanotubes: Science and Applications* (CRC Press, Boca Raton, FL, 2004).
- [29] O. V. Motygin and S. A. Nazarov, A computer-aided procedure for constructing boundary layers in plate theory, *Comp. Math. Math. Phys.* **40**(2), 261–272 (2000).
- [30] E. Oberg, F. D. Jones, Ch. J. McCauley, and M. Ricardo, *Machinery's Handbook*, 27th ed. (Industrial Press, New York, 2004).
- [31] G. P. Panasenko, *Multi-Scale Modelling for Structures and Composites* (Springer-Verlag, Berlin, 2005).
- [32] E. Sanchez-Palencia, *Non-Homogeneous Media and Vibration Theory* (Springer-Verlag, Berlin, 1980).
- [33] S. Timoshenko and J. N. Goodier, *Theory of Elasticity* (McGraw-Hill, New York, 1951).
- [34] L. Trabuco and J. M. Viaño, Mathematical Modeling of Rods, In: *Handbook of Numerical Analysis*, Vol. IV, edited by P. G. Ciarlet and J.-L. Lions, (North-Holland, Amsterdam, 1996), pp. 487–974.
- [35] S. P. Timoshenko, *History of Strength of Materials* (Dover, New York, 1983).
- [36] I. Titeux and E. Sanchez-Palencia, Junction of thin plates, *Eur. J. Mech. A/Solids* **19**, 377400 (2000).
- [37] www.ansys.com, ANSYS 6.0 User Manual, ANSYS Inc. (2001).